

Chapter 5

Sequences and Series

Solutions

SECTION - A

Objective Type Questions (One option is correct)]

[Arithmetic Progression]

1. If general term of a sequence is $n(n+1)(2n+1)$, then its 5th term is
(1) 110 (2) 155 (3) 330 (4) 420

Sol. Answer (3)

$$\text{Here, } a_n = n(n+1)(2n+1)$$

$$\therefore a_5 = 5 \times 6 \times 11 = 330$$

2. If general term of an A.P. is $2n+5$, then its common difference is
(1) 2 (2) 3 (3) 5 (4) 7

Sol. Answer (1)

$$\text{Here, } a_n = 2n+5$$

$$\therefore a_1 = 7$$

$$a_2 = 9$$

$$\text{Thus, } d = a_2 - a_1 = 2$$

Alternate method:

$$a_n = 2n+5$$

$$a_{n-1} = 2(n-1) + 5 = 2n+3$$

$$\therefore d = a_n - a_{n-1} = 2$$

3. If 6th and 12th term of an A.P. are 13 and 25 respectively, then its 20th term is
(1) 37 (2) 39 (3) 41 (4) 43

Sol. Answer (3)

$$\text{Here } a_6 = a + 5d = 13$$

$$\text{and } a_{12} = a + 11d = 25$$

$$\therefore 6d = 12$$

$$\Rightarrow d = 2$$

$$\text{and } a = 3$$

$$\text{Hence } a_{20} = a + 19d = 3 + 38 = 41$$

4. If a, b, c, d, e, f are in A.P., then $e - c$ is equal to

(1) $d - c$

(2) $2(d - c)$

(3) $2(c - a)$

(4) $c - b$

Sol. Answer (2)

Let D be the common difference,

$$\text{then } c = a + 2D$$

$$d = a + 3D$$

$$e = a + 4D$$

$$\text{Clearly, } e - c = 2D = 2(d - c)$$

5. The m^{th} term of an A.P. is n and n^{th} term is m . Its p^{th} term is

(1) $m - n + p$

(2) $n + p - m$

(3) $m + n - p$

(4) $m + n + p$

Sol. Answer (3)

$$\text{Given, } a_m = n \Rightarrow a + (m - 1)d = n \quad \dots(i)$$

$$a_n = m \Rightarrow a + (n - 1)d = m \quad \dots(ii)$$

Subtracting (ii) from (i), we get

$$(m - n)d = n - m$$

$$\Rightarrow d = -1$$

$$\text{and } a = n + m - 1 = (m + n - 1)$$

$$\therefore a_p = a + (p - 1)d$$

$$= m + n - 1 + 1 - p = m + n - p$$

6. If 8th term of an A.P. is 15, then the sum of first 15 terms is

(1) 180

(2) 210

(3) 225

(4) 240

Sol. Answer (3)

$$\text{Given, } a_8 = 15 \Rightarrow a + 7d = 15$$

$$\therefore S_{15} = \frac{15}{2}[2a + (15 - 1)d]$$

$$= \frac{15}{2}(2a + 14d)$$

$$= 15(a + 7d)$$

$$= 15 \times 15$$

$$= 225$$

7. If $a_1, a_2, a_3, \dots$ are in A.P. such that $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$, then $a_1 + a_2 + a_3 + \dots + a_{24}$ is equal to

(1) 600

(2) 900

(3) 1200

(4) 1800

Sol. Answer (2)

$$3(a_1 + a_{24}) = 225$$

$$\Rightarrow a_1 + a_{24} = 75$$

$$\text{Required sum} = \frac{24}{2}(a_1 + a_{24}) = 900$$

8. The first and last terms of an A.P. are 1 and 7. If the sum of its terms is 36, then the number of terms will be

(1) 6

(2) 7

(3) 8

(4) 9

Sol. Answer (4)

$$\text{Given, } a = 1, l = 7 \text{ and } S_n = 36$$

$$\therefore S_n = \frac{n}{2}[a+l]$$

$$\therefore 36 = \frac{n}{2}[1+7]$$

$$\Rightarrow n = \frac{72}{8} = 9$$

9. An A.P. has common difference 2, sum of first n terms 49 and 7th term 13. The value of n is equal to

(1) 5

(2) 6

(3) 7

(4) 8

Sol. Answer (3)

$$\text{Given } d = 2, S_n = 49, a_7 = 13$$

$$\therefore a + 6d = 13$$

$$\Rightarrow a = 1$$

$$\text{Also, } 49 = \frac{n}{2}[2a + (n-1)d]$$

$$\Rightarrow 49 = \frac{n}{2}[2 + (n-1) \times 2]$$

$$\Rightarrow 9 = n(1 + n - 1)$$

$$\Rightarrow n^2 = 49$$

$$\Rightarrow n = 7$$

10. If the sum of first n terms of an A.P. is $2n^2 + 5n$, then its n^{th} term is

(1) $3n - 5$

(2) $4n - 3$

(3) $4n + 3$

(4) $3n + 5$

Sol. Answer (3)

$$S_n = 2n^2 + 5n$$

$$\therefore S_1 = a = 7$$

$$\therefore S_2 = 8 + 10 = 18 = 2a + d$$

$$\Rightarrow d = 4$$

$$T_n = a + (n - 1)d$$

$$= 7 + (n - 1)4$$

$$= 4n + 3$$

11. The sum of all two digit odd numbers is

(1) 2375

(2) 2475

(3) 2560

(4) 4920

Sol. Answer (2)

Sequence of two digit odd numbers : 11, 13, 15, ... 99

$$\text{No. of terms } n = \frac{99-11}{2} + 1 = 45$$

$$\therefore \text{Required sum} = \frac{45}{2}(11+99) \\ = 45 \times 55 = 2475$$

12. The number of numbers lying between 81 and 1792 which are divisible by 17 is

(1) 105

(2) 107

(3) 109

(4) 101

Sol. Answer (4)

Numbers divisible by 17 and lying between 81 and 1792 are 85, 102, 119, ..., 1785

$$\therefore 1785 = 85 + (n - 1) \times 17$$

$$\Rightarrow 1700 = (n - 1) \times 17$$

$$\Rightarrow 100 = n - 1$$

$$\Rightarrow n = 101$$

13. Three numbers are in A.P. such that their sum is 24 and sum of their squares is 200. The numbers are

(1) 2, 8, 14

(2) 4, 8, 12

(3) 6, 8, 10

(4) 5, 8, 11

Sol. Answer (3)

$$(a - d) + a + (a + d) = 24$$

$$\Rightarrow a = 8$$

$$\text{Also, } (8 - d)^2 + 8^2 + (8 + d)^2 = 200$$

$$\Rightarrow 64 - 16d + d^2 + 64 + 64 + d^2 + 16d = 200$$

$$\Rightarrow 2d^2 + 192 = 200$$

$$\Rightarrow 2d^2 = 8$$

$$\Rightarrow d = \pm 2$$

Hence numbers are 6, 8, 10

14. The sum to p terms of an A.P. is q and the sum to q terms is p . The sum to $p + q$ terms is

- (1) $-(p + q)$ (2) 0 (3) $p - q$ (4) $p + q$

Sol. Answer (1)

$$\frac{p}{2}[2a + (p-1)d] = q \quad \dots (i)$$

$$\frac{q}{2}[2a + (q-1)d] = p \quad \dots (ii)$$

Equation (i) – Equation (ii)

$$\Rightarrow a(p-q) + \frac{d}{2}[(p^2 - q^2) - (p-q)] = q - p$$

$$\Rightarrow a + \frac{d}{2}(p+q-1) = -1$$

15. If $\frac{3+5+7+\dots \text{to } n \text{ terms}}{5+8+11+\dots \text{to } 10 \text{ terms}} = 7$, then n is equal to

- (1) 35 (2) 36 (3) 37 (4) 38

Sol. Answer (1)

$$\frac{\frac{n}{2}[2 \times 3 + (n-1) \times 2]}{\frac{10}{2}[2 \times 5 + (9 \times 3)]} = 7$$

$$\Rightarrow \frac{n}{2}(6+2n-2) = 7 \times 5 \times 37$$

$$\Rightarrow 2n^2 + 4n = 2590$$

$$\Rightarrow n^2 + 2n - 1295 = 0$$

$$\Rightarrow n^2 + 37n - 35n - 1295 = 0$$

$$\Rightarrow n(n+37) - 35(n+37) = 0$$

$$\Rightarrow (n+37)(n-35) = 0$$

$$\therefore n = 35 \quad (n \neq -37)$$

16. If n A.M.'s are inserted between 3 and 17 such that the ratio of the last mean to the first mean is 3 : 1, then the value of n is

- (1) 4 (2) 6 (3) 8 (4) 9

Sol. Answer (2)

$$\text{Given, } \frac{3 + \left(\frac{17-3}{n+1}\right)n}{3 + \left(\frac{17-3}{n+1}\right)} = \frac{3}{1}$$

$$\Rightarrow 9 + \frac{42}{n+1} = 3 + \frac{14n}{n+1}$$

$$\Rightarrow 51 + 9n = 17n + 3$$

$$\Rightarrow 8n = 48$$

$$\Rightarrow n = 6$$

17. Four numbers are in A.P. The sum of first and last is 8 and the product of both middle terms is 15. The least among the four numbers is

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (1)

Let, $a - 3d, a - d, a + d, a + 3d$ be the four numbers.

Given, $a - 3d + a + 3d = 2a = 8$

$$\Rightarrow a = 4$$

Also, $a^2 - d^2 = 15$

$$\Rightarrow 16 - d^2 = 15$$

$$\Rightarrow d^2 = 1$$

$$\Rightarrow d = \pm 1. \text{ Hence numbers are } 1, 3, 5, 7.$$

18. If the sum of three numbers in A.P is 27 and the product of first and last is 77, then the numbers are

(1) $\frac{7}{2}, \frac{51}{4}, 22$

(2) 6, 9, 12

(3) 7, 10, 11

(4) 7, 9, 11

Sol. Answer (4)

Let $a - d, a, a + d$ be the numbers

then $3a = 27$

$$\Rightarrow a = 9$$

and $a^2 - d^2 = 77$

$$\Rightarrow 81 - d^2 = 77$$

$$\Rightarrow d = \pm 2$$

Hence numbers are 7, 9, 11.

19. If n A.M.'s are inserted between 2 and 38 such that the sum of the resulting series obtained is 200, then the value of n is

(1) 6

(2) 8

(3) 10

(4) 12

Sol. Answer (2)

$$\text{Sum of } n \text{ A.M.s.} = n \left(\frac{a+b}{2} \right)$$

$$\text{Sum of series} = n \left(\frac{a+b}{2} \right) + a + b = 200$$

$$\Rightarrow (n+2)(a+b) = 400$$

$$\Rightarrow n+2 = \frac{400}{40} = 10$$

$$\Rightarrow n = 8$$

20. If S_n denotes the sum to n terms of an A.P. whose first term is a and common difference is d , then $S_{n+3} - 3S_{n+2} + 3S_{n+1} - S_n$ is equal to

- (1) $-d$ (2) $-a$ (3) 0 (4) $2a$

Sol. Answer (3)

$$\begin{aligned} & S_{n+3} - 3S_{n+2} + 3S_{n+1} - S_n \\ &= \frac{n+3}{2} [2a + (n+3-1)d] - 3 \frac{(n+2)}{2} [2a + (n+2-1)d] + 3 \frac{(n+1)}{2} [2a + (n+1-1)d] - \frac{n}{2} [2a + (n-1)d] \\ &= 2a \left(\frac{n+3}{2} - \frac{3n+6}{2} + \frac{3n+3}{2} - \frac{n}{2} \right) + d \left[\frac{(n+2)(n+3)}{2} - \frac{3(n+2)(n+1)}{2} + \frac{3n(n+1)}{2} - \frac{n(n-1)}{2} \right] \\ &= 0 \end{aligned}$$

21. If three positive numbers a, b, c are in A.P. and $\frac{1}{a^2}, \frac{1}{b^2}, \frac{1}{c^2}$ also in A.P., then

- (1) $a = b = c$ (2) $2b = 3a + c$ (3) $b^2 = \frac{ac}{8}$ (4) $2c = 2b + a$

Sol. Answer (1)

$$\begin{aligned} & \frac{2}{b^2} = \frac{1}{a^2} + \frac{1}{c^2} \\ \Rightarrow & \frac{2}{(a+c)^2} = \frac{(a+c)^2 - 2ac}{a^2c^2} \\ \Rightarrow & (a+c)^2 = 4ac, -2ac \\ \Rightarrow & a = c \text{ and } (a+c)^2 \neq -2ac \end{aligned}$$

22. If S_n denotes the sum of first n terms of an A.P., and $S_{2n} = 3S_n$ then $\frac{S_{3n}}{S_n}$ is equal to

- (1) 4 (2) 5 (3) 6 (4) 7

Sol. Answer (3)

$$\begin{aligned} & \frac{2n}{2} [2a + (2n-1)d] = \frac{3n}{2} [2a + (n-1)d] \\ \Rightarrow & 2a = (n+1)d \end{aligned}$$

23. Let S_n denotes the sum to n terms of an A.P. whose first term is a . If the common difference d is given by $d = S_n - kS_{n-1} + S_{n-2}$, then k is equal to

- (1) 1 (2) 2 (3) 3 (4) 4

Sol. Answer (2)

24. If the first, second and last term of an A.P. are a , b and $2a$ respectively, then its sum is

(1) $\frac{ab}{2(b-a)}$

(2) $\frac{ab}{b-a}$

(3) $\frac{3ab}{2(b-a)}$

(4) $\frac{2ab}{a-b}$

Sol. Answer (3)

$$d = b - a$$

Let n be the number of terms

$$\text{then } 2a = a + (n - 1)(b - a)$$

$$\frac{a}{b-a} = n - 1$$

$$n = \frac{a}{b-a} + 1 = \frac{b}{b-a}$$

$$\therefore \text{Sum} = \frac{b}{2(b-a)}(a+2a) = \frac{3ab}{2(b-a)}$$

25. If A_1, A_2 are two A.M.'s between two positive numbers a and b , then $(2A_1 - A_2)(2A_2 - A_1)$ is equal to

(1) $a + b$

(2) ab

(3) $\frac{ab}{a+b}$

(4) $\frac{a}{b}$

Sol. Answer (2)

$$A_1 = a + \frac{b-a}{3}$$

$$A_2 = a + 2\left(\frac{b-a}{3}\right)$$

$$\begin{aligned} \text{Now, } (2A_1 - A_2)(2A_2 - A_1) &= \left[2a + \frac{2(b-a)}{3} - a - \frac{2(b-a)}{3}\right] \left[2a + \frac{4(b-a)}{3} - a - \frac{b-a}{3}\right] \\ &= a(a + b - a) \\ &= ab \end{aligned}$$

26. If tenth term of an A.P. is 19 and sum of first fifteen terms is 225 then fifth term of A.P. is

(1) 5

(2) 6

(3) 9

(4) 11

Sol. Answer (3)

Let first term is a and common difference is d , then

$$T_{10} = a + 9d = 19$$

$$S_{15} = \frac{15}{2}[2a + (15-1) \times d] = 225$$

$$2a + 14d = 30 \quad \dots (i)$$

$$a + 9d = 19 \quad \dots (ii)$$

On solving equation (i) & (ii) we get, $a = 1$ and $d = 2$ then $T_5 = a + (5-1)d = 1 + 4 \times 2 = 9$

27. The maximum sum of the series $100 + 98 + 96 + \dots$ is

- (1) 2500 (2) 2550 (3) 2050 (4) 2555

Sol. Answer (2)

For maximum sum,

$$T_n = a + (n-1)d \geq 0$$

$$\Rightarrow 100 + (n-1) \times (-2) \geq 0$$

$$\Rightarrow n-1 \leq 50 \Rightarrow n \leq 51$$

$$\text{Then, } S_n = \frac{51}{2}[100 + 0] = 51 \times 50 = 2550$$

28. Consider that 10 arithmetic means are inserted between 3 and 7 and their sum is a . Again consider that the sum of five numbers in A.P. is 30 and the value of middle terms is b . Then $a + b$ equals

- (1) 16 (2) 56 (3) 46 (4) 36

Sol. Answer (2)

Let, $A_1, A_2, A_3, \dots, A_{10}$ are 10 A.M.'s between 3 and 7

$$\text{then } A_1 + A_2 + A_3 + \dots + A_{10} = 10 \left(\frac{3+7}{2} \right) = 50$$

Now sum of five terms of an A.P.

$$b - 2d + b - d + b + b + d + b - 2d = 30$$

$$\Rightarrow 5b = 30$$

$$\Rightarrow b = 6$$

$$a + b = 50 + 6 = 56$$

29. Let t_r denote the r th term of an A.P. Also suppose $t_m = \frac{1}{n}$ and $t_n = \frac{1}{m}$, ($m \neq n$), for some positive integers m and n , then which of the following is necessarily a root of the equation ?

$$(l+m-2n)x^2 + (m+n-2l)x + (n+l-2m)=0$$

- (1) t_n (2) t_m (3) t_{m+n} (4) t_{mn}

Sol. Answer (4)Let first terms of A.P. = a Common difference = d

$$t_m = \frac{1}{n} = a + (m-1)d \quad \dots(i)$$

$$t_n = \frac{1}{m} = a + (n-1)d \quad \dots(ii)$$

by (i) and (ii)

$$a = \frac{1}{mn}, d = \frac{1}{mn}$$

$$t_{mn} = a + (mn-1)d$$

$$= \frac{1}{mn} + (mn-1) \times \frac{1}{mn} = \frac{1+mn-1}{mn} = 1$$

Also $x=1$, is the root of the equation, hence root is t_{mn} .

30. The sum of the first 100 terms common to the series 17, 21, 25, 29, 33, and 16, 21, 26, 31, 36..... is

(1) 100101

(2) 111000

(3) 101100

(4) 110010

Sol. Answer (3)Common difference of first A.P. = $d_1 = 4$ Common difference of second A.P. = $d_2 = 5$ L.C.M. of $(d_1, d_2) = 20$

First common terms of both A.P. is 21

Hence the A.P. formed by common terms is

21, $21 + 20$, $21 + 2 \times 20$, $21 + 3 \times 20$,

$$\begin{aligned} \text{Sum of 100 terms} &= \frac{100}{2} [2 \times 21 + (100-1) \times 20] \\ &= 50 [42 + 99 \times 20] = 50 [42 + 1980] \\ &= 50 [2022] = 101100 \end{aligned}$$

[Geometric Progression]

31. The fourth term of the G.P. 4, -2, 1, ... is

(1) $-\frac{1}{2}$ (2) $\frac{1}{2}$

(3) -1

(4) $\frac{1}{4}$ **Sol.** Answer (1)

$$a_4 = ar^3 = (4) \times \left(\frac{-1}{2}\right)^3 = \frac{-4}{8} = -\frac{1}{2}$$

32. Which term of the G.P. $18, -12, 8, \dots$ is $\frac{512}{729}$?

(1) 7^{th} (2) 9^{th} (3) 11^{th} (4) 13^{th}

Sol. Answer (2)

$$a_n = ar^{n-1}$$

$$\Rightarrow \frac{512}{729} = 18 \times \left(\frac{-2}{3}\right)^{n-1}$$

$$\Rightarrow \left(\frac{-2}{3}\right)^{n-1} = \frac{512}{729 \times 18} = \frac{256}{81 \times 9 \times 9} = \left(\frac{-2}{3}\right)^8$$

$$\Rightarrow n - 1 = 8$$

$$\Rightarrow n = 9$$

33. The third term of a G.P. is 3. The product of its first five terms is

(1) 81

(2) 243

(3) 729

(4) 343

Sol. Answer (2)

$$ar^2 = 3$$

$$\text{Now, } a \cdot ar \cdot ar^2 \cdot ar^3 \cdot ar^4 = a^5 r^{10} = (ar^2)^5 = 3^5 = 243$$

34. If $(2p)^{\text{th}}$ term of a G.P. is q^2 and $(2q)^{\text{th}}$ term is p^2 , then $(p+q)^{\text{th}}$ term is

(1) pq (2) $\frac{1}{2}p^2q^2$ (3) p^2q^2 (4) $\frac{1}{4}p^3q^3$

Sol. Answer (1)

$$ar^{2p-1} = q^2 \quad \dots (i)$$

$$ar^{2q-1} = p^2 \quad \dots (ii)$$

Multiply (i) and (ii),

$$ar^{p+q-1} = pq$$

35. If first and eighth terms of a G.P. are x^{-4} and x^{52} and its second term is x^t , then t is equal to

(1) 2

(2) 3

(3) 4

(4) 13

Sol. Answer (3)

$$a = x^{-4}$$

$$\text{and } ar^7 = x^{52} \Rightarrow (x^{-4})r^7 = x^{52}$$

$$\Rightarrow r^7 = x^{56}$$

$$\Rightarrow r = x^8$$

$$\therefore ar = x^{-4} = x^t$$

$$\therefore t = 4$$

36. The sum to n terms of the G.P. $1 + \frac{1}{2} + \frac{1}{4} + \dots$ is

(1) $\frac{1}{2^n}$

(2) $\frac{1}{2^n} + 1$

(3) $1 - \frac{1}{2^n}$

(4) $2\left(1 - \frac{1}{2^n}\right)$

Sol. Answer (4)

$$S_n = \frac{1\left[1 - \left(\frac{1}{2}\right)^n\right]}{1 - \frac{1}{2}} = 2\left(1 - \frac{1}{2^n}\right)$$

37. The n^{th} term of a G.P. is 128 and the sum of its n terms is 255. If its common ratio is 2, then its first term is

(1) 1

(2) 3

(3) 5

(4) 7

Sol. Answer (1)

$$ar^{n-1} = 128$$

$$\text{and } \frac{a(r^n - 1)}{r - 1} = 255$$

$$\Rightarrow \frac{ar^{n-1} \cdot r - a}{r - 1} = 255$$

$$\Rightarrow \frac{128 \times 2 - a}{1} = 255$$

$$\Rightarrow 256 - 255 = a$$

$$\Rightarrow a = 1$$

$$\therefore a = 1$$

38. The sum of three numbers in a G.P. is 26 and the sum of products taken two at a time is 156, then the numbers are

(1) 1, 5, 25

(2) 2, 6, 18

(3) 1, 4, 16

(4) 2, 8, 16

Sol. Answer (2)

$$a + ar + ar^2 = 26 \quad \dots (i)$$

$$a^2r(1 + r + r^2) = 156 \quad \dots (ii)$$

Squaring (i) and divide by (ii),

$$r = 3, \frac{1}{3} \text{ and } a = 2, 18$$

39. If a, b, c are in G.P., then $\frac{b-a}{b-c} + \frac{b+a}{b+c}$ is equal to

(1) ab

(2) ac

(3) $c^2 - b^2$

(4) 0

Sol. Answer (4)

Since a, b, c are in G.P.

$$b = ar, c = ar^2$$

$$\begin{aligned}\text{Now, } \frac{b-a}{b-c} + \frac{b+a}{b+c} &= \frac{a(r-1)}{ar(1-r)} + \frac{a(r+1)}{ar(1+r)} \\ &= \frac{1}{r} - \frac{1}{r} = 0\end{aligned}$$

40. If x , $2x + 2$, $3x + 3$ are the first three terms of G.P., then the fourth term is

- (1) -27 (2) 27 (3) -13.5 (4) 13.5

Sol. Answer (3)

$$(2x + 2)^2 = x(3x + 3)$$

$$\Rightarrow x^2 + 5x + 4 = 0$$

$$\Rightarrow x = -1, -4$$

$$\text{But } x \neq -1, x = -4$$

41. If g_1, g_2, g_3 are three geometric means between two positive numbers a and b , then $g_1 g_3$ is equal to

- (1) g_2 (2) $2g_2$ (3) g_2^2 (4) g_2^3

Sol. Answer (3)

g_1, g_2, g_3 are also in G.P.

$$\therefore g_1 g_3 = g_2^2$$

42. The fifth term of a G.P. is 32 and common ratio is 2, then the sum of first 14 terms of the G.P. is

- (1) 16388 (2) 32667 (3) 32766 (4) 64342

Sol. Answer (3)

$$a \times 2^4 = 32 \Rightarrow a = 2$$

$$\therefore S_{14} = \frac{2(r^{14} - 1)}{r - 1} = \frac{2(2^{14} - 1)}{1} = 32766$$

43. If the sum of first three numbers in G.P. is 21 and their product is 216, then the numbers are

- (1) 3, 6, 12 (2) 5, 7, 9 (3) 6, 2, 213 (4) 6, 12, 24

Sol. Answer (1)

$$\frac{a}{r} \cdot a \cdot ar = 216$$

$$\Rightarrow a^3 = 216$$

$$\Rightarrow a = 6$$

$$\text{Also, } \frac{6}{r} + 6 + 6r = 21$$

$$\Rightarrow 6 \left(\frac{r^2 + 1}{r} \right) = 15$$

$$\Rightarrow 2r^2 + 2 - 5r = 0$$

$$\Rightarrow 2r^2 - 4r - r + 2 = 0$$

$$\Rightarrow 2r(r - 2) - (r - 2) = 0$$

$$\Rightarrow r = \frac{1}{2}, 2$$

Hence numbers are 3, 6, 12

44. If x, y, z are three geometric means between 6 and 54, then z is equal to

(1) $9\sqrt{3}$

(2) 18

(3) $18\sqrt{3}$

(4) 27

Sol. Answer (3)

$$z = 3rd \text{ G.M.} = ar^3 = a\left(\frac{b}{a}\right)^{\frac{3}{n+1}} = 6\left(\frac{54}{6}\right)^{\frac{3}{4}} = 6 \times 3\sqrt{3} = 18\sqrt{3}$$

45. If three distinct numbers a, b, c are in A.P. and $b - a, c - b, a$ are in G.P., then $a : b : c$ is equal to

(1) $2 : 3 : 4$

(2) $1 : 2 : 4$

(3) $1 : 2 : 3$

(4) $1 : 3 : 5$

Sol. Answer (3)

$$b - a = c - b$$

$$2b = a + c$$

$$\text{Also, } (c - b)^2 = a(b - a)$$

$$\Rightarrow c - b = a$$

$$\Rightarrow a + b = c$$

$$\Rightarrow a + \frac{a+c}{2} = c$$

$$\Rightarrow 3a + c = 2c$$

$$\Rightarrow 3a = c$$

$$\Rightarrow \frac{a}{c} = \frac{1}{3}$$

$$2b = a + 3a = 4a$$

$$\therefore \frac{a}{b} = \frac{1}{2}$$

Hence required ratio = $1 : 2 : 3$

46. Three numbers form an increasing G.P. If the middle term is doubled, then the numbers are in A.P. The common ratio of the G.P. is

(1) $2 - \sqrt{3}$

(2) $2 + \sqrt{3}$

(3) $\sqrt{3} - 2$

(4) $3 + \sqrt{2}$

Sol. Answer (2)

$$4ar = a + ar^2$$

$$\Rightarrow r = 2 \pm \sqrt{3}$$

For an increasing G.P.,

$$r = 2 + \sqrt{3}$$

47. The sum $1 + 3 + 3^2 + \dots + 3^n$ is equal to

- (1) 3^{n+1} (2) $\frac{3^{n+1}-1}{2}$ (3) $3^{n+1} - 1$ (4) 3^n

Sol. Answer (2)

$$1 + 3 + 3^2 + \dots + 3^n$$

$$\therefore S_n = \frac{3^{n+1}-1}{2}$$

48. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$ to ∞ is equal to

- (1) $\frac{1}{2}$ (2) 1 (3) 2 (4) 3

Sol. Answer (2)

$$S_{\infty} = \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$$

49. If the sixth term of a GP be 2, then the product of first eleven terms is

- (1) 1024 (2) 2047 (3) 2048 (4) 1023

Sol. Answer (3)

Let G.P. is $a, ar, ar^2, ar^3, \dots$

$$T_6 = ar^5 = 2 \quad \dots (i)$$

$$P = a \cdot ar \cdot ar^2 \dots (ar^9) (ar^{10})$$

$$= a^{11} r^{1+2+3+\dots+10}$$

$$= a^{11} r^{\frac{10 \times 11}{2}}$$

$$= a^{11} r^{55}$$

$$= (ar^5)^{11} = 2^{11}$$

$$= 2048 \text{ by (i)}$$

50. Let a_n be the n^{th} term of the G.P. of positive numbers. Let $\sum_{n=1}^{100} a_{2n} = \alpha$ and $\sum_{n=1}^{100} a_{2n-1} = \beta$ such that $\alpha \neq \beta$, then the common ratio is

(1) $\frac{\alpha}{\beta}$

(2) $\frac{\beta}{\alpha}$

(3) $\sqrt{\frac{\alpha}{\beta}}$

(4) $\sqrt{\frac{\beta}{\alpha}}$

Sol. Answer (1)

Let G.P. = $a, ar, ar^2, \dots$, where ' r ' is the common ratio

$$\alpha = a_2 + a_4 + \dots + a_{200} = ar + ar^3 + ar^5 + \dots + ar^{199}$$

$$\alpha = \frac{ar(1-(r^2)^{100})}{1-r^2} = \frac{ar(1-r^{200})}{(1-r^2)} \quad \dots(i)$$

$$\text{Similarly, } \beta = a_1 + a_3 + a_5 + \dots + a_{199}$$

$$= a + ar^2 + ar^4 + \dots + ar^{198}$$

$$\beta = \frac{a(1-(r^2)^{100})}{(1-r^2)} = \frac{a(1-r^{200})}{(1-r^2)} \quad \dots(ii)$$

By (i) and (ii)

$$\frac{\alpha}{\beta} = r$$

51. The series $\frac{2x}{x+3} + \left(\frac{2x}{x+3}\right)^2 + \left(\frac{2x}{x+3}\right)^3 + \dots \infty$ will have a definite sum when

(1) $-1 < x < 3$

(2) $0 < x < 1$

(3) $x = 0$

(4) $x = -3$

Sol. Answer (1)

For definite sum

$$-1 < \frac{2x}{x+3} < 1$$

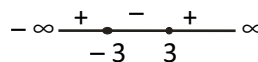
$$\text{Case (i) } \frac{2x}{x+3} < 1 \Rightarrow \frac{2x}{x+3} - 1 < 0$$

$$\frac{2x - (x+3)}{x+3} < 0$$

$$\Rightarrow \frac{x-3}{x+3} < 0 \Rightarrow \frac{(x-3)(x+3)}{(x+3)^2} < 0$$

$$\Rightarrow (x-3)(x+3) < 0$$

$$\Rightarrow x \in (-3, 3) \quad \dots(i)$$



Case (ii) $\frac{2x}{x+3} > -1 \Rightarrow \frac{2x}{x+3} + 1 > 0$

$$\Rightarrow \frac{2x+x+3}{x+3} > 0$$

$$\Rightarrow \frac{3x+3}{x+3} > 0 \Rightarrow \frac{(x+1)}{(x+3)} > 0 \Rightarrow -\infty \overset{+}{-3} \overset{-}{-1} \overset{+}{\infty}$$

$$\Rightarrow \frac{(x+1)(x+3)}{(x+3)^2} > 0$$

$$\text{or } (x+1)(x+3) > 0$$

$$x \in (-\infty, -3) \cup (-1, \infty) \quad \dots \text{(ii)}$$

By (i) and (ii) the common interval of x is $(-1, 3)$

52. Four geometric means are inserted between the numbers $2^{11} - 1$ and $2^{11} + 1$. The product of these geometric means is

(1) $2^{22} - 2^{11} + 1$

(2) $2^{44} - 2^{23} + 1$

(3) $2^{44} - 2^{22} + 1$

(4) $2^{22} - 2^{12} + 1$

Sol. Answer (2)

The product of Geometric mean

$$= \left(\sqrt{(2^{11}-1)(2^{11}+1)} \right)^4 = (2^{22}-1)^2 = 2^{44} - 2 \cdot 2^{22} + 1 = 2^{44} - 2^{23} + 1$$

53. The value of x in $(-\pi, \pi)$ which satisfies the equation $8^{1+|\cos x|+|\cos^2 x|+|\cos^3 x|+\dots\infty} = 4^3$ is

(1) $\frac{\pi}{3}$

(2) $-\frac{\pi}{3}$

(3) $\frac{2\pi}{3}$

(4) All of these

Sol. Answer (4)

$$8^{1+|\cos x|+|\cos^2 x|+\dots\infty}$$

$$= 8^{1+|\cos x|+|\cos^2 x|+|\cos^3 x|+\dots\infty} = 8^{\frac{1}{1-|\cos x|}} = 4^3 = 8^2$$

$$\Rightarrow 1-|\cos x| = \frac{1}{2}$$

$$|\cos x| = \frac{1}{2}$$

$$\cos x = \pm \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{3}, -\frac{\pi}{3}, \frac{2\pi}{3}, -\frac{2\pi}{3}, \text{ hence only four solutions exist}$$

54. If a, b, c are three distinct real numbers and they are in a G.P. If $a + b + c = xb$, then

(1) $x < -1$ or $x > 3$

(2) $x \leq -3$ or $x \geq 2$

(3) $x \leq -4$ or $x \geq 3$

(4) $x \leq 2$ or $x \geq 5$

Sol. Answer (1)

Let common ratio = r

$$\begin{array}{c} + \quad - \quad + \\ -1 \quad 3 \end{array}$$

$$\Rightarrow b = ar, c = ar^2$$

$$a + b + c = xb$$

$$\Rightarrow a + ar + ar^2 = x(ar)$$

$$\Rightarrow 1 + r + r^2 = xr$$

$$r^2 + r(1 - x) + 1 = 0$$

Because r is real, hence Discriminant > 0

$$\Rightarrow (1 - x)^2 - 4 \times 1 \times (1) > 0$$

$$((1 - x) - 2)(1 - x + 2) > 0$$

$$(-1 - x)(-x + 3) > 0$$

$$(x + 1)(x - 3) > 0$$

$$\Rightarrow x \in (-\infty, -1) \cup (3, \infty)$$

$$\begin{array}{c} + \quad - \quad + \\ -1 \quad 3 \end{array}$$

55. The product $2^{\frac{1}{4}} \cdot 4^{\frac{1}{8}} \cdot 8^{\frac{1}{16}} \dots$ to ∞ equal to

(1) 4

(2) $\sqrt{2}$

(3) 8

(4) 2

Sol. Answer (4)

$$2^{\frac{1}{4}} \cdot 4^{\frac{1}{8}} \cdot 8^{\frac{1}{16}} \cdot 16^{\frac{1}{32}} \dots$$

$$= 2^{\frac{1}{4}} \cdot 2^{\frac{2}{8}} \cdot 2^{\frac{3}{16}} \cdot 2^{\frac{4}{32}} \dots$$

$$= 2^{\frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \dots}$$

$$\text{Let } P = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \dots \dots \dots$$

...(i)

$$\Rightarrow \frac{P}{2} = \frac{1}{8} + \frac{2}{16} + \frac{3}{32} + \dots \dots \dots$$

...(ii)

Subtracting (ii) from (i)

$$P - \frac{P}{2} = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \dots \dots \text{ upto } \infty$$

$$\frac{P}{2} = \frac{\frac{1}{4}}{1 - \frac{1}{2}} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

$$P = 1$$

$$\Rightarrow 2^{\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots} = 2^P = 2^1 = 2$$

[Mixed Problems of A.P. and G. P.]

56. Three numbers whose product is 512 are in G.P. If 8 is added to the first and 6 to the second, the number will be in A.P. The numbers are

- (1) 2, 8, 32 (2) 8, 8, 8 (3) 4, 8, 16 (4) 2, 8, 14

Sol. Answer (3)

$$\text{Given, } \frac{a}{r} \cdot a \cdot ar = 512$$

$$\Rightarrow a^3 = 512$$

$$\Rightarrow a = 8$$

$$\text{Now, } \frac{8}{r} + 8, 14, 8r \text{ are in A.P.}$$

$$\Rightarrow 8r + 8\left(\frac{1}{r} + 1\right) = 28$$

$$\Rightarrow \frac{8r^2 + 8 + 8r}{r} = 28$$

$$\Rightarrow 8(r^2 + r + 1) = 28r$$

$$\Rightarrow 2r^2 - 5r + 2 = 0$$

$$\Rightarrow 2r^2 - 4r - r + 2 = 0$$

$$\Rightarrow 2r(r - 2) - (r - 2) = 0$$

$$\Rightarrow 2r = 1 \text{ or } r = 2$$

$$\Rightarrow r = \frac{1}{2}, 2$$

Hence numbers are 4, 8, 16.

57. If a, b, c form a G.P. with common ratio r such that $0 < r < 1$, and if $a, \frac{3}{2}b, -4c$ form an A.P., then r is equal to

- (1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) $\frac{1}{4}$

Sol. Answer (4)

$$3ar = a - ar^2$$

$$\Rightarrow r = -1, \frac{1}{4}$$

$$\text{But } 0 < r < 1, r = \frac{1}{4}$$

58. If x, y, z are in A.P.; ax, by, cz are in G.P. and $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P., then $\frac{x}{z} + \frac{z}{x}$ is equal to

- (1) $\frac{a}{b} + \frac{b}{a}$ (2) $\frac{a}{c} + \frac{c}{a}$ (3) $\frac{a}{c} + \frac{c}{b}$ (4) $\frac{b}{c} + \frac{c}{a}$

Sol. Answer (2)

$$b^2y^2 = ac \times z$$

$$\Rightarrow \frac{4a^2c^2}{(a+c)^2} \cdot \frac{(x+z)^2}{4} = ac \times z$$

$$\Rightarrow \frac{(x+z)^2}{xz} = \frac{(a+c)^2}{ac}$$

$$\Rightarrow \frac{x}{z} + \frac{z}{x} = \frac{a}{c} + \frac{c}{a}$$

59. If the second, third and sixth terms of an A.P. are distinct and form consecutive terms of a G.P., then the common ratio of the G.P. is

(1) $\frac{1}{2}$

(2) 1

(3) 2

(4) 3

Sol. Answer (4)

$a + d, a + 2d, a + 5d$ are in G.P.

$$\Rightarrow \frac{a+5d}{a+2d} = \frac{a+2d}{a+d}$$

$$\Rightarrow a^2 + 6ad + 5d^2 = a^2 + 4d^2 + 4ad$$

$$\Rightarrow d^2 + 2ad = 0$$

$$\Rightarrow d(d + 2a) = 0$$

$$\Rightarrow d \neq 0, \therefore d = -2a$$

Now, $a + d = -a$

$$a + 2d = -3a$$

$$a + 5d = -9a$$

common ratio = 3

60. If distinct positive numbers a, b, c are in G.P. and $\frac{1}{a-b}, \frac{1}{c-a}, \frac{1}{b-c}$ are in A.P., then the value of $a + 4b + c$ is equal to

(1) -3

(2) 0

(3) 3

(4) 4

Sol. Answer (2)

$$\frac{2}{ar^2 - a} = \frac{1}{a - ar} + \frac{1}{ar - ar^2}$$

$$\Rightarrow \frac{-2}{r+1} = 1 + \frac{1}{r}$$

$$\Rightarrow r^2 + 2r + 1 = -2r$$

$$\Rightarrow ar^2 + 4ar + a = 0$$

or $a + 4b + c = 0$

61. If one geometric mean G and two A.Ms p and q be inserted between two given quantities then $(2p - q)(2q - p)$ equals

(1) G (2) G^2 (3) $2G^2$ (4) $2G$

Sol. Answer (2)

Let the numbers are a and b .

$$\Rightarrow G^2 = ab \quad \dots(i)$$

Also a, p, q, b be in A.P.

$$\Rightarrow 2p = a + q \Rightarrow (2p - q) = a$$

$$\text{and } 2q = p + b \Rightarrow (2q - p) = b$$

$$\Rightarrow (2p - q)(2q - p) = ab = G^2 \text{ by (i)}$$

62. The sum of three numbers in G.P. is 56. If we subtract 1, 7, 21 from these numbers in that order, we obtain an A.P. the three numbers are

(1) 10, 18, 26 (2) 8, 16, 32 (3) 9, 16, 23 (4) 5, 8, 11

Sol. Answer (2)

Let the numbers are a, ar, ar^2

$$a + ar + ar^2 = 56 \quad \dots(i)$$

$a - 1, ar - 7, ar^2 - 21$ are in A.P.

$$\Rightarrow 2(ar - 7) = (a - 1) + (ar^2 - 21)$$

$$2ar - 14 = a + ar^2 - 22 \quad \dots(ii)$$

By (i) and (ii)

$$2ar - 14 = (56 - ar) - 22$$

$$3ar = 56 - 22 + 14$$

$$3ar = 48$$

$$ar = 16 \quad \dots(iii)$$

By (i), (iii)

$$a + 16 + ar^2 = 56$$

$$a + ar^2 = 40$$

$$\text{Again } ar = 16 \Rightarrow a = \frac{16}{r}$$

$$\Rightarrow \frac{16}{r} + \frac{16}{r} \cdot r^2 = 40$$

solving, we get

$$r = 2, r = \frac{1}{2}$$

Case (i) if $r = 2$, then $a = 8$

then numbers are 8, 16, 32.

Case (ii) if $r = \frac{1}{2}$ then $a = 32$

then numbers are 32, 16, 8.

63. The sum of the series $1 + 2.2 + 3.2^2 + 4.2^3 + 5.2^4 + \dots + 100.2^{99}$ is

(1) $99.2^{100} - 1$

(2) 100.2^{100}

(3) 99.2^{100}

(4) $99.2^{100} + 1$

Sol. Answer (4)

$$S = 1 + 2.2 + 3.2^2 + 4.2^3 + \dots + 100.2^{99} \quad \dots(i)$$

$$2S = 2 + 2.2^2 + 3.2^3 + \dots + 99.2^{99} + 100.2^{100} \quad \dots(ii)$$

Subtracting (ii) from (i)

$$-S = 1 + (2.2 - 2) + (3.2^2 - 2.2^2) + \dots + (100.2^{99} - 99.2^{99}) - 100.2^{100}$$

$$-S = 1 + 2 + 2^2 + 2^3 + \dots + 2^{99} - 100.2^{100}$$

$$-S = \frac{2^{100} - 1}{2 - 1} - 100.2^{100}$$

$$-S = 2^{100} - 1 - 100.2^{100}$$

$$S = 99.2^{100} + 1$$

64. If $3 + \frac{1}{4}(3+p) + \frac{1}{4^2}(3+2p) + \dots$ to $\infty = 8$, then p equals

(1) 1

(2) 5

(3) 7

(4) 9

Sol. Answer (4)

$$3 + \frac{1}{4}(3+p) + \frac{1}{4^2}(3+2p) + \dots = \text{to } \infty = 8$$

$$\text{Let } S = 3 + \frac{1}{4}(3+p) + \frac{1}{4^2}(3+2p) + \dots \quad \dots(i)$$

$$\Rightarrow \frac{S}{4} = \frac{1}{4}(3) + \frac{1}{4^2}(3+p) + \dots \quad \dots(ii)$$

Subtracting (ii) from (i)

$$\frac{3S}{4} = 3 + \frac{1}{4}(p) + \frac{1}{4^2}(p) + \dots$$

$$\frac{3S}{4} = 3 + p \left(\frac{1}{4} + \frac{1}{4^2} + \dots \right)$$

$$\frac{3S}{4} = 3 + p \left(\frac{\frac{1}{4}}{1 - \frac{1}{4}} \right)$$

$$\frac{3S}{4} = 3 + \frac{p}{3}$$

But $S = 8$

$$\Rightarrow \frac{3 \times 8}{4} = 3 + \frac{p}{3}$$

$$\Rightarrow \frac{p}{3} = 6 - 3 = 3 \Rightarrow p = 9$$

65. Let S denotes the infinite sum

$$2 + 5x + 9x^2 + 14x^3 + 20x^4 + \dots,$$

where $|x| < 1$. Then S equals

(1) $\frac{2-x}{(1+x)^3}$

(2) $\frac{2-x}{(1-x)^3}$

(3) $\frac{2+x}{(1-x)^3}$

(4) $\frac{2+x}{(1+x)^3}$

Sol. Answer (2)

$$S = 2 + 5x + 9x^2 + 14x^3 + 20x^4 + \dots \quad \dots(i)$$

$$\Rightarrow Sx = 2x + 5x^2 + 9x^3 + 14x^4 + \dots \quad \dots(ii)$$

Subtracting (ii) from (i)

$$S(1-x) = 2 + 3x + 4x^2 + 5x^3 + 6x^4 + \dots \quad \dots(iii)$$

$$S(1-x)x = 2x + 3x^2 + 4x^3 + 5x^4 + \dots \quad \dots(iv)$$

Subtracting (iv) from (iii)

$$S(1-x)(1-x) = 2 + x + x^2 + x^3 + \dots$$

$$S(1-x)^2 = 2 + \frac{x}{1-x}$$

$$S(1-x)^2 = \frac{2-2x+x}{(1-x)} = \frac{2-x}{(1-x)}$$

$$S = \frac{2-x}{(1-x)^3}$$

66. The coefficient of x^{101} in the expansion of

$$(1-x)(1-2x)(1-2^2x) \dots (1-2^{101}x) \text{ is}$$

(1) $2^{4950} - 2^{5050}$

(2) $2^{5051} - 2^{5152}$

(3) $2^{4950} - 2^{5051}$

(4) $2^{5050} - 2^{5152}$

Sol. Answer (4)

$$(1-x)(1-2x)(1-2^2x)(1-2^3x) \dots (1-2^{101}x)$$

$$= (-1)(x-1)(-2)\left(x-\frac{1}{2}\right)(-2)^2\left(x-\frac{1}{2^2}\right)(-2)^3\left(x-\frac{2}{2^3}\right) \dots (-2^{101})\left(x-\frac{1}{2^{101}}\right)$$

$$= (-1)^{102} 2 \cdot 2^2 \cdot 2^3 \cdot 2^4 \dots 2^{101} (x-1)\left(x-\frac{1}{2}\right)\left(x-\frac{1}{2^2}\right) \dots \left(x-\frac{1}{2^{101}}\right)$$

$$= 2^{1+2+3+\dots+101} \cdot (x-1)\left(x-\frac{1}{2}\right)\left(x-\frac{1}{2^2}\right) \dots \left(x-\frac{1}{2^{101}}\right)$$

$$= 2^{\frac{101 \times 102}{2}} \cdot (x^{102} - \left(1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^{101}}\right) x^{101} - \dots)$$

Coefficient of x^{101}

$$\begin{aligned}
 &= 2^{\frac{101 \times 102}{2}} (-1) \left(1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{101}} \right) \\
 &= 2^{101 \times 51} (-1) \left(\frac{\left(\frac{1}{2}\right)^{102} - 1}{\frac{1}{2} - 1} \right) = 2^{101 \times 51} (-1) \left(\frac{1 - \left(\frac{1}{2}\right)^{102}}{1 - \frac{1}{2}} \right) \\
 &= -2^{101 \times 51} \cdot \left(\frac{2^{102} - 1}{2^{102}} \right) \times \frac{2}{1} \\
 &= -2^{101 \times 51} \cdot 2^{-101} (2^{102} - 1) \\
 &= 2^{5050} - 2^{5152}
 \end{aligned}$$

[Special Series]67. The sum of the series $1^2 + 1 + 2^2 + 2 + 3^2 + 3 + \dots + n^2 + n$ is equal to

(1) $\frac{n(n+1)(n+2)}{3}$ (2) $\frac{n(n+1)(3n+1)}{2}$ (3) $\frac{n(2n+1)(3n+1)}{6}$ (4) $\left[\frac{n(n+1)}{3} \right]^2$

Sol. Answer (1)

$$(1 + 2 + 3 + \dots + n) + (1^2 + 2^2 + \dots + n^2)$$

$$= \frac{n(n+1)}{2} + \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{n(n+1)}{2} \left[1 + \frac{2n+1}{3} \right]$$

$$= \frac{n(n+1)}{2} \left(\frac{3+2n+1}{3} \right)$$

$$= \frac{n(n+1)(n+2)}{3}$$

68. The sum of the series $3 + 7 + 13 + 21 + 31 + \dots$ to n terms is equal to

(1) $\frac{n(n^2+2n+3)}{3}$ (2) $\frac{n(n^2+3n+5)}{3}$ (3) $\frac{n(n^2+2n+5)}{3}$ (4) $\frac{n(n^2+2n+2)}{3}$

Sol. Answer (2)

$$\text{Let } S_n = 3 + 7 + 13 + 21 + \dots + a_{n-1} + a_n \quad \dots(i)$$

$$\text{Again } S_n = 3 + 7 + 13 + \dots + a_{n-2} + a_{n-1} + a_n \quad \dots(ii)$$

Subtracting (ii) from (i),

$$0 = 3 + [4 + 6 + 8 + \dots + (a_n - a_{n-1})] - a_n$$

$$\Rightarrow a_n = 3 + \frac{(n-1)}{2} [2 \times 4 + (n-1-1) \times 2]$$

$$\Rightarrow a_n = 3 + \frac{(n-1)}{2} (8 + 2n - 4)$$

$$= 3 + \frac{(n-1)}{2} (2n + 4)$$

$$= 3 + n^2 + n - 2 = n^2 + n + 1$$

$$\therefore S_n = \sum (n^2 + n + 1) = \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + n = \frac{n(n^2 + 3n + 5)}{3}$$

69. The sum of the series $\frac{7}{1^2 \cdot 6^2} + \frac{17}{6^2 \cdot 11^2} + \frac{27}{11^2 \cdot 16^2} + \dots$ is

- (1) 1 (2) $\frac{1}{6}$ (3) $\frac{1}{5}$ (4) $\frac{1}{4}$

Sol. Answer (3)

$$5S = \frac{5 \cdot 7}{1^2 \cdot 6^2} + \frac{5 \cdot 17}{6^2 \cdot 11^2} + \frac{17 \cdot 27}{11^2 \cdot 16^2} + \dots$$

$$= \frac{6^2 - 1^2}{1^2 \cdot 6^2} + \frac{11^2 - 6^2}{6^2 \cdot 11^2} + \frac{16^2 - 11^2}{11^2 \cdot 16^2} + \dots$$

$$= 1 - \frac{1}{6^2} + \frac{1}{6^2} - \frac{1}{11^2} + \frac{1}{11^2} - \frac{1}{16^2} + \frac{1}{16^2} - \dots = 1$$

$$\Rightarrow S = \frac{1}{5}$$

70. The sum to 100 terms of the series

1.2.3. + 2.3.4. + 3.4.5. + ... + $n(n+1)(n+2)$ + ... is integral multiple of

- (1) 2525 (2) 2526 (3) 2527 (4) 2528

Sol. Answer (1)

$$T_n = n(n+1)(n+2) = n^3 + 3n^2 + 2n$$

$$S_n = \sum T_n = \sum n^3 + 3\sum n^2 + 2\sum n$$

$$= \left(\frac{n(n+1)}{2} \right)^2 + 3 \frac{n(n+1)(2n+1)}{6} + \frac{2n(n+1)}{2}$$

$$= \frac{n(n+1)(n+2)(n+3)}{4}$$

$$\Rightarrow S_{100} = \frac{100 \times 101 \times 102 \times 103}{4} = 2525 \times \text{Integer}$$

71. The sum of the first n terms of the series $1^2 + 2.2^2 + 3^2 + 2.4^2 + \dots$ is $\frac{n(n+1)^2}{2}$ when n is even. Then the sum if n is odd, is

- (1) $\frac{n(n^2+1)}{3}$ (2) $n(n+1)^2$ (3) $\frac{n^2(n+1)}{2}$ (4) $\frac{n(n^2+1)}{2}$

Sol. Answer (3)If n is odd then $n - 1$ is even then

$$S_{n-1} = \frac{(n-1)n^2}{2}, t_n = n^2$$

$$\Rightarrow S_n = S_{n-1} + t_n = \frac{(n-1)n^2}{2} + n^2 = \frac{(n+1)n^2}{2}$$

72. If x, y, z and w are non-zero real numbers and $x^2 + 5y^2 + 5z^2 + 4w^2 - 4xy - 4yz - 4zw = 0$, then x, y, z, w are in

(1) A.P.

(2) A.G.P.

(3) H.P.

(4) G.P.

Sol. Answer (4)

The given expression can be written as

$$(x - 2y)^2 + (y - 2z)^2 + (z - 2w)^2 = 0$$

$$x = 2y, y = 2z, z = 2w$$

$$\frac{x}{y} = \frac{y}{z} = \frac{z}{w} = 2$$

Hence x, y, z, w are in G.P.

[Properties of Logarithmic Function, Logarithmic Expansion and Exponential Series]

73. Let $\log_2 3 = \alpha$, then $\log_{64} 729$ is

(1) α (2) $\frac{\alpha}{3}$ (3) 3α (4) 2α **Sol.** Answer (1)

$$\log_2 3 = \alpha$$

$$\log_{64} 729 = \log_{(2^6)} (3^6) = \frac{6}{6} \log_2 3 = \alpha$$

74. $\sum_{n=0}^{\infty} \frac{(\ln x)^n}{n!}$ is equal to

(1) $\ln x$ (2) x (3) $\frac{1}{\ln x}$ (4) $\frac{1}{x}$ **Sol.** Answer (2)

$$\sum_{n=0}^{\infty} \frac{(\ln x)^n}{n!} = 1 + \frac{(\ln x)}{1!} + \frac{(\ln x)^2}{2!} + \dots = e^{\ln x} = x$$

75. The sum of $\sum_{n=2}^{\infty} {}^nC_2 \cdot \frac{3^{n-2}}{n!}$ equals

(1) $\frac{e}{2}$ (2) e^2 (3) e^3 (4) $\frac{1}{2}e^3$

Sol. Answer (4)

$$\sum_{n=2}^{\infty} {}^nC_2 \cdot \frac{3^{n-2}}{n!} = \sum_{n=2}^{\infty} \frac{n(n-1)}{2!} \cdot \frac{3^{n-2}}{n(n-1)(n-2)!}$$

$$\frac{1}{2} \sum_{n=2}^{\infty} \frac{3^{n-2}}{(n-2)!} = \frac{1}{2} \left[\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!} + \dots \right] = \frac{e^3}{2}$$

76. The coefficient of x^4 in $\frac{1-3x+x^2}{e^x}$ equals

(1) $\frac{12}{15}$

(2) $\frac{24}{25}$

(3) $\frac{25}{24}$

(4) $\frac{15}{12}$

Sol. Answer (3)

$$\frac{(1-3x+x^2)}{e^x} = (1-3x+x^2)e^{-x}$$

$$= (1-3x+x^2) \left(1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots \right)$$

$$\text{Coefficient of } x^4 = \frac{1}{4!} + \frac{3}{3!} + \frac{1}{2!} = \frac{25}{24}$$

77. The sum of the series

$$\frac{2}{1!} + \frac{4}{3!} + \frac{6}{5!} + \dots \text{ to } \infty \text{ equals}$$

(1) $e + 1$

(2) $e - 1$

(3) e

(4) e^{-1}

Sol. Answer (3)

$$\frac{2}{1!} + \frac{4}{3!} + \frac{6}{5!} + \dots \infty$$

$$t_r = \frac{2r}{(2r-1)!} = \frac{2r-1+1}{(2r-1)!} = \frac{1}{(2r-2)!} + \frac{1}{(2r-1)!}$$

$$S = \sum_{r=1}^{\infty} t_r = \sum_{r=1}^{\infty} \frac{1}{(2r-2)!} + \sum_{r=1}^{\infty} \frac{1}{(2r-1)!}$$

$$= \left(1 + \frac{1}{2!} + \frac{1}{4!} + \dots \right) + \left(\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \dots \right)$$

$$= e$$

78. $\frac{2}{1!} + \frac{2+4}{2!} + \frac{2+4+6}{3!} + \frac{2+4+6+8}{4!} + \dots \text{ to } \infty \text{ equals}$

(1) e

(2) $2e$

(3) $3e$

(4) $3e - 2$

Sol. Answer (3)

$$\frac{2}{1!} + \frac{2+4}{2!} + \frac{2+4+6}{3!} + \frac{2+4+6+8}{4!} + \dots \infty$$

$$t_r = \frac{2+4+6+\dots+2r}{r!} = \frac{2(1+2+3+\dots+r)}{r!} = \frac{r(r+1)}{r!}$$

$$\Rightarrow t_r = \frac{r+1}{(r-1)!} = \frac{r-1+2}{(r-1)!} = \frac{1}{(r-2)!} + \frac{2}{(r-1)!}$$

$$S = \sum_{r=1}^{\infty} t_r = \sum_{r=1}^{\infty} \frac{1}{(r-2)!} + 2 \sum_{r=1}^{\infty} \frac{1}{(r-1)!}$$

$$\Rightarrow S = \left(1 + \frac{1}{1!} + \frac{1}{2!} + \dots\right) + 2 \left(1 + \frac{1}{1!} + \frac{1}{2!} + \dots\right) = e + 2e = 3e$$

$$79. \frac{\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots}{\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \dots} =$$

$$(1) \frac{1-e}{1+e}$$

$$(2) \frac{e-1}{e+1}$$

$$(3) \frac{e+1}{1-e}$$

$$(4) \frac{e+1}{e-1}$$

Sol. Answer (2)

$$\frac{\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots}{\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \dots} = \frac{\frac{e+e^{-1}}{2} - 1}{\frac{e-e^{-1}}{2}} = \frac{e^2 - 2e + 1}{e^2 - 1} = \frac{(e-1)^2}{(e+1)(e-1)} = \frac{e-1}{e+1}$$

80. The sum of the series

$$(a+b)(a-b) + \frac{(a+b)(a-b)(a^2+b^2)}{2!} + \frac{(a+b)(a-b)(a^4+b^4+a^2b^2)}{3!} + \dots \text{ to } \infty$$

is equal to

$$(1) e^{a^2} + e^{b^2}$$

$$(2) e^{a^2+b^2}$$

$$(3) e^{a^2-b^2}$$

$$(4) e^{a^2} - e^{b^2}$$

Sol. Answer (4)

$$\begin{aligned} & (a+b)(a-b) + \frac{(a+b)(a-b)(a^2+b^2)}{2!} + \dots \\ &= a^2 - b^2 + \frac{a^4 - b^4}{2!} + \frac{a^6 - b^6}{3!} + \dots \\ &= \left(a^2 + \frac{a^4}{2!} + \frac{a^6}{3!} + \dots\right) - \left(b^2 + \frac{b^4}{2!} + \frac{b^6}{3!} + \dots\right) \\ &= e^{a^2} - 1 - (e^{b^2} - 1) = e^{a^2} - e^{b^2} \end{aligned}$$

81. The sum of the series $\left(\frac{a-b}{a}\right) + \frac{1}{2}\left(\frac{a-b}{a}\right)^2 + \frac{1}{3}\left(\frac{a-b}{a}\right)^3 + \dots \infty$ will be

- (1) $\log_e ab$ (2) $\log_e \left(\frac{b}{a}\right)$ (3) $\log_e \left(\frac{a}{b}\right)$ (4) $\log_e a^b$

Sol. Answer (3)

$$\begin{aligned} & \left(\frac{a-b}{a}\right) + \frac{1}{2}\left(\frac{a-b}{a}\right)^2 + \frac{1}{3}\left(\frac{a-b}{a}\right)^3 + \dots \\ &= -\log_e \left[1 - \left(\frac{a-b}{a}\right)\right] = -\log_e \frac{b}{a} = \log_e \frac{a}{b} \end{aligned}$$

82. The sum of series $\frac{1}{1.2} - \frac{1}{2.3} + \frac{1}{3.4} - \frac{1}{4.5} + \dots$ is

- (1) $2 \log_e 2$ (2) $\log_e 2 - 1$ (3) $\log_e 2$ (4) $\log_e \left(\frac{4}{e}\right)$

Sol. Answer (4)

$$\begin{aligned} & \frac{1}{1.2} - \frac{1}{2.3} + \frac{1}{3.4} - \frac{1}{4.5} + \dots \\ &= 1 - \frac{1}{2} - \frac{1}{2} + \frac{1}{3} + \frac{1}{3} - \frac{1}{4} - \frac{1}{4} + \frac{1}{5} + \dots \\ &= 1 - 2\left(\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \dots\right) \\ &= 1 - 2(1 - \log_e 2) = 2\log_e 2 - 1 \\ &= \log_e \left(\frac{4}{e}\right) \end{aligned}$$

83. $\sum_{n=0}^{\infty} \frac{n^2 + n + 1}{(n+1)!}$ is equal to

- (1) $2e$ (2) $2e + 1$ (3) $2e - 1$ (4) $1 - 2e$

Sol. Answer (3)

$$\sum_{n=0}^{\infty} \frac{n^2 + n + 1}{(n+1)!} = \sum_{n=0}^{\infty} \frac{n(n+1)}{(n+1)!} + \frac{1}{(n+1)!} = \sum_{n=0}^{\infty} \frac{1}{(n-1)!} + \sum_{n=0}^{\infty} \frac{1}{(n+1)!} = e + e - 1 = 2e - 1$$

84. If $x = \frac{1}{1.2} + \frac{1}{3.4} + \frac{1}{5.6} + \dots \infty$, and

$$y = 1 - \frac{1}{2.3} - \frac{1}{4.5} - \frac{1}{6.7} - \dots \infty, \text{ then}$$

- (1) $x = y$ (2) $x > y$ (3) $x < y$ (4) $xy = 1$

Sol. Answer (1)

$$x = \frac{1}{1.2} + \frac{1}{3.4} + \frac{1}{5.6} + \dots \infty$$

$$\Rightarrow x = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

$$y = 1 - \frac{1}{2.3} - \frac{1}{4.5} - \frac{1}{6.7} - \dots$$

$$\Rightarrow y = 1 - \left(\frac{1}{2} - \frac{1}{3}\right) - \left(\frac{1}{4} - \frac{1}{5}\right) - \left(\frac{1}{6} - \frac{1}{7}\right) - \dots = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$$

$$\Rightarrow x = y$$

85. If $x^2y = 2x - y$ and $|x| < 1$, then $\frac{\left(y + \frac{y^3}{3} + \frac{y^5}{5} + \dots \infty\right)}{\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \infty\right)} =$

(1) 1

(2) -1

(3) 2

(4) -2

Sol. Answer (3)

$$\Rightarrow \frac{y + \frac{y^3}{3} + \frac{y^5}{5} + \dots \infty}{x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \infty} = \frac{\log_e \left(\frac{1+y}{1-y}\right)}{\log_e \left(\frac{1+x}{1-x}\right)}$$

$$\Rightarrow x^2y = 2x - y \Rightarrow y = \frac{2x}{x^2 + 1}$$

$$\Rightarrow 1 + y = 1 + \frac{2x}{x^2 + 1} = \frac{x^2 + 1 + 2x}{x^2 + 1} = \frac{(x+1)^2}{x^2 + 1}$$

$$\Rightarrow 1 - y = 1 - \frac{2x}{x^2 + 1} = \frac{x^2 + 1 - 2x}{x^2 + 1} = \frac{(1-x)^2}{x^2 + 1}$$

$$\Rightarrow \frac{1+y}{1-y} = \left(\frac{1+x}{1-x}\right)^2 \Rightarrow \frac{\log_e \left(\frac{1+y}{1-y}\right)}{\log_e \left(\frac{1+x}{1-x}\right)} = \frac{\log_e \left(\frac{1+x}{1-x}\right)^2}{\log_e \left(\frac{1+x}{1-x}\right)} = 2$$

86. $2 \left[\frac{1}{2x+1} + \frac{1}{3(2x+1)^3} + \frac{1}{5(2x+1)^5} + \dots \right]$ is equal to

(1) $\log\left(\frac{x}{x+1}\right)$

(2) $\log\left(\frac{x+1}{x}\right)$

(3) $\log(2x+1)$

(4) $\log\left(\frac{1}{2x+1}\right)$

Sol. Answer (2)

$$\therefore \log\left(\frac{1+x}{1-x}\right) = 2 \left\{ x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \right\}$$

$$2 \left[\frac{1}{2x+1} + \frac{1}{3(2x+1)^3} + \frac{1}{5(2x+1)^5} + \dots \right]$$

$$\log \left(\frac{1 + \frac{1}{2x+1}}{1 - \frac{1}{2x+1}} \right) = \log \left(\frac{2x+2}{2x} \right) = \log \left(\frac{x+1}{x} \right)$$

87. If $\frac{e^x}{1-x} = B_0 + B_1x + B_2x^2 + \dots + B_nx^n + \dots$, then $B_n - B_{n-1}$ equals;

- (1) $\frac{1}{n!}$ (2) $\frac{1}{(n-1)!}$ (3) $\frac{1}{n!} - \frac{1}{(n-1)!}$ (4) 1

Sol. Answer (1)

$$\frac{e^x}{1-x} = B_0 + B_1x + B_2x^2 + \dots + B_nx^n + \dots$$

$$\Rightarrow e^x(1-x)^{-1} = B_0 + B_1x + B_2x^2 + \dots + B_nx^n + \dots$$

$$\therefore \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots \right) \left(1 + x + x^2 + \dots + x^n + \dots \right) = B_0 + B_1x + B_2x^2 + \dots + B_nx^n + \dots$$

On equating coefficients of x^n and x^{n-1} on both sides, we get

$$\frac{1}{n!} + \frac{1}{(n-1)!} + \dots + \frac{1}{2!} + \frac{1}{1!} + 1 = B_n$$

$$\text{and } \frac{1}{(n-1)!} + \frac{1}{(n-2)!} + \dots + \frac{1}{2!} + \frac{1}{1!} + 1 = B_{n-1} \Rightarrow B_n - B_{n-1} = \frac{1}{n!}$$

[Miscellaneous]

88. If $\theta_1, \theta_2, \theta_3, \dots, \theta_n$ are in A.P. whose common difference is d , then $\sin d [\sec \theta_1 \sec \theta_2 + \sec \theta_2 \sec \theta_3 + \dots + \sec \theta_{n-1} \sec \theta_n] =$

- (1) $\sec \theta_n - \sec \theta_1$ (2) $\sin \theta_n - \sin \theta_1$ (3) $\cos \theta_n - \cos \theta_1$ (4) $\tan \theta_n - \tan \theta_1$

Sol. Answer (4)

$$\theta_2 - \theta_1 = \theta_3 - \theta_2 = \dots = \theta_n - \theta_{n-1} = d$$

$$\Rightarrow \sin d [\sec \theta_1 \sec \theta_2 + \sec \theta_2 \sec \theta_3 + \dots + \sec \theta_{n-1} \sec \theta_n]$$

$$= \frac{\sin d}{\cos \theta_1 \cos \theta_2} + \frac{\sin d}{\cos \theta_2 \cos \theta_3} + \dots + \frac{\sin d}{\cos \theta_n \cos \theta_{n-1}}$$

$$= \frac{\sin(\theta_2 - \theta_1)}{\cos \theta_2 \cos \theta_1} + \frac{\sin(\theta_3 - \theta_2)}{\cos \theta_3 \cos \theta_2} + \dots + \frac{\sin(\theta_n - \theta_{n-1})}{\cos \theta_n \cos \theta_{n-1}}$$

$$= \frac{\sin \theta_2 \cos \theta_1 - \cos \theta_2 \sin \theta_1}{\cos \theta_2 \cos \theta_1} + \frac{\sin \theta_3 \cos \theta_2 - \cos \theta_3 \sin \theta_2}{\cos \theta_3 \cos \theta_2} + \dots$$

$$\Rightarrow (\tan \theta_2 - \tan \theta_1) + (\tan \theta_3 - \tan \theta_2) + \dots + (\tan \theta_n - \tan \theta_{n-1})$$

$$\Rightarrow \tan \theta_n - \tan \theta_1, \text{ remaining terms are cancelled out}$$

89. Let a, b be the roots of the equation $x^2 - 4x + k_1 = 0$ and c, d the roots of the equation $x^2 - 36x + k_2 = 0$. If $a < b < c < d$ and a, b, c, d are in GP, then the product $k_1 k_2$ equals
- (1) 81 (2) 729 (3) 256 (4) 64

Sol. Answer (2)

Let common ratio of G.P. is r

$$b = ar, c = ar^2, d = ar^3$$

$$x^2 - 4x + k_1 = 0$$

By the properties of roots

$$a + b = 4 \Rightarrow a + ar = 4 \quad \dots(i)$$

$$ab = k_1 \Rightarrow a(ar) = k_1$$

$$\Rightarrow a^2 r = k_1 \quad \dots(ii)$$

Similarly,

$$x^2 - 36x + k_2 = 0$$

$$ar^2 + ar^3 = 36 \quad \dots(iii)$$

$$(ar^2) \times (ar^3) = k_2 \quad \dots(iv)$$

By (i) and (iii),

$$\frac{a(1+r)}{r^2 a(1+r)} = \frac{4}{36} \Rightarrow \frac{1}{r^2} = \frac{1}{9} \Rightarrow r = \pm 3$$

Case (i) if $r = 3$, then by (i) $a = 1$

$$k_1 = 3, k_2 = 243$$

$$\Rightarrow k_1 k_2 = 243 \times 3 = 729$$

By $a < b < c < d$ we find that r is positive hence only one case is possible. That is given by $r = 3$

90. The value of $x + y + z$ is 15 if a, x, y, z, b are in A.P. while the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ is $\frac{5}{3}$ if a, x, y, z, b are in H.P. the value of a and b are

- (1) 9, 1 (2) 7, 4 (3) 8, 2 (4) -1, 3

Sol. Answer (1)

We know that the sum of n A.M. between two quantities are equal to n times their single mean.

$$\Rightarrow x + y + z = 3 \left(\frac{a+b}{2} \right)$$

$$\Rightarrow 15 = \frac{3}{2} (a + b) \Rightarrow a + b = 10 \quad \dots(i)$$

Similarly a, x, y, z, b are in H.P.

$$\Rightarrow \frac{1}{a}, \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{b} \text{ are in A.P.}$$

$$\text{Again } \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{3}{2} \left(\frac{1}{a} + \frac{1}{b} \right)$$

$$\Rightarrow \frac{5}{3} = \frac{3}{2} \left(\frac{1}{a} + \frac{1}{b} \right)$$

$$\Rightarrow \frac{10}{9} = \frac{a+b}{ab} = \frac{10}{ab}$$

$$\Rightarrow ab = 9$$

...(ii)

By (i) and (ii) we get that $a(10 - a) = 9$

$$\Rightarrow a^2 - 10a + 9 = 0$$

$$(a - 1)(a - 9) = 0 \Rightarrow a = 1, 9$$

if $a = 1$ then $b = 9$ and if $a = 9$, then $b = 1$

Hence numbers are 9, 1

91. If x, y, z are positive reals satisfying $4xy + 6yz + 8zx = 9$, then the greatest possible value of the product xyz is

(1) 0.125

(2) 0.375

(3) 0.500

(4) 0.625

Sol. Answer (2)

Using A.M. $\geq$ G.M.

$$\frac{4xy + 6yz + 8zx}{3} \geq ((4xy)(6yz)(8zx))^{\frac{1}{3}}$$

$$\Rightarrow \frac{9}{3} \geq (4 \times 4 \times 2 \times 2 \times 3 \times x^2 y^2 z^2)^{\frac{1}{3}}$$

$$\Rightarrow 27 \geq 192 x^2 y^2 z^2$$

$$\Rightarrow x^2 y^2 z^2 \leq \frac{27}{192}$$

$$xyz \leq 0.375$$

92. If $a + b + c = 3$ and $a > 0, b > 0, c > 0$, then the greatest value of $a^2 b^3 c^2$ is

(1) $\frac{3^9 \cdot 2^4}{7^7}$

(2) $\frac{3^{10} \cdot 2^4}{7^7}$

(3) $\frac{3^8 \cdot 2^4}{7^7}$

(4) $\frac{3^7 \cdot 2^4}{7^7}$

Sol. Answer (2)

$A \geq G$

$$\frac{\frac{a}{2} + \frac{a}{2} + \frac{b}{3} + \frac{b}{3} + \frac{b}{3} + \frac{c}{2} + \frac{c}{2}}{7} \geq \left\{ \left(\frac{a}{2} \right)^2 \left(\frac{b}{3} \right)^3 \left(\frac{c}{2} \right)^2 \right\}^{\frac{1}{7}}$$

$$\Rightarrow \left(\frac{3}{7} \right)^7 \geq \frac{a^2}{2^2} \cdot \frac{b^3}{3^3} \cdot \frac{c^2}{2^2}$$

$$\frac{3^{10} \cdot 2^4}{7^7} \geq a^2 b^2 c^3$$

$$\Rightarrow a^2 b^2 c^3 \leq \frac{3^{10} \cdot 2^4}{7^7}$$

Hence the maximum value is $\frac{3^{10} \cdot 2^4}{7^7}$.

93. Let C be a circle with centre P_0 and AB be a diameter of C . Suppose P_1 is the midpoint of line segment P_0B , P_2 the midpoint of line segment P_1B and so on. Let C_1, C_2, C_3 be circles with diameters $P_0P_1, P_1P_2, P_2P_3, \dots$ respectively. Suppose the circles $C_1, C_2, C_3 \dots$ are all shaded. The ratio of the area of the unshaded portion of C to that of the original circle is

(1) 8 : 9

(2) 9 : 10

(3) 10 : 11

(4) 11 : 12

Sol. Answer (4)

Let the radius of largest circle is $r \Rightarrow AB = 2r$

$$P_0P_1 = \frac{r}{2}, P_0C_1 = \frac{r}{4}$$

$$\text{Area of circle } C_1 = \pi \left(\frac{r}{4} \right)^2 = \frac{\pi r^2}{4^2}$$

$$\text{Similarly area of circle } C_2 = \pi \left(\frac{r}{8} \right)^2 = \frac{\pi r^2}{8^2}$$

$$\text{Area of the circle } C_3 = \pi \left(\frac{r}{16} \right)^2 = \frac{\pi r^2}{16^2} \text{ and so on.}$$

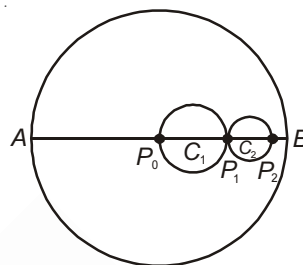
The sum of all areas is

$$= \pi r^2 \left[\frac{1}{4^2} + \frac{1}{8^2} + \frac{1}{16^2} + \dots \dots \dots \infty \right]$$

$$S = \pi r^2 \left[\frac{\frac{1}{16}}{1 - \frac{1}{4}} \right] = \pi r^2 \left[\frac{\frac{1}{16}}{\frac{3}{4}} \right] = \frac{\pi r^2}{12}$$

$$S_1 = \text{Area of unshaded portion} = \pi r^2 - \frac{\pi r^2}{12} = \frac{11\pi r^2}{12}$$

$$\frac{S_1}{\pi r^2} = \frac{\frac{11}{12}\pi r^2}{\pi r^2} = \frac{11}{12}$$



94. If $1, \log_9(3^{1-x} + 2)$ and $\log_3(4 \cdot 3^x - 1)$ are in A.P., then x is

(1) $\log_3 4$

(2) $1 - \log_3 4$

(3) $\log_3 0.25$

(4) $\log_4 3$

Sol. Answer (2)

We have,

$$2\log_9(3^{1-x} + 2) = 1 + \log_3(4 \cdot 3^x - 1)$$

$$\Rightarrow \log_3(3^{1-x} + 2) = \log_3 3(4 \cdot 3^x - 1)$$

$$\Rightarrow 3^{1-x} + 2 = 12 \cdot 3^x - 3$$

$$\Rightarrow 3 + 2 \cdot 3^x = 12 \cdot (3^x)^2 - 3 \cdot 3^x$$

$$\Rightarrow 12 \cdot (3^x)^2 - 5 \cdot 3^x - 3 = 0$$

$$\Rightarrow 12y^2 - 5y - 3 = 0, \text{ let } 3^x = y$$

$$\Rightarrow 12y^2 - 9y + 4y - 3 = 0$$

$$\Rightarrow 3y(4y - 3) + 1(4y - 3) = 0$$

$$\Rightarrow y = -\frac{1}{3}, \frac{3}{4}$$

$$3^x = \frac{3}{4}$$

$$\Rightarrow x \log 3 = \log 3/4$$

$$\Rightarrow x = \log_3 3 - \log_3 4$$

$$\Rightarrow 1 - \log_3 4$$

95. If $b - c$, $2b - x$ and $b - a$ are in HP. Then $a - \left(\frac{x}{2}\right)$, $b - \left(\frac{x}{2}\right)$ and $c - \left(\frac{x}{2}\right)$ are in

(1) AP

(2) GP

(3) HP

(4) None of these

Sol. Answer (2)

$b - c$, $2b - x$, $b - a$ are in HP

$$\Rightarrow \frac{1}{b-c}, \frac{1}{2b-x}, \frac{1}{b-a} \text{ are in AP}$$

$$\Rightarrow \frac{2}{2b-x} = \frac{1}{b-c} + \frac{1}{b-a}$$

$$\Rightarrow 2(b-c)(b-a) = (2b-x)(2b-a-c)$$

$$\Rightarrow 2(b^2 - ab - bc + ac) = 4b^2 - 2ab - 2bc - 2bx + ax + cx$$

$$\Rightarrow 2ac = 2b^2 - 2bx + ax + cx$$

$$\Rightarrow 2ac - \frac{x}{2}(a+c) = b^2 - xb$$

$$\Rightarrow ac - \frac{x}{2}(a+c) + \frac{x^2}{4} = b^2 - xb + \frac{x^2}{4}$$

$$\Rightarrow \left(a - \frac{x}{2}\right)\left(c - \frac{x}{2}\right) = \left(b - \frac{x}{2}\right)^2$$

$$\Rightarrow a - \frac{x}{2}, b - \frac{x}{2}, c - \frac{x}{2} \text{ are in GP.}$$

96. The numbers $\log_{180} 12$, $\log_{2160} 12$, $\log_{25920} 12$ are in

(1) A.P.

(2) G.P.

(3) H.P.

(4) None of the above progressions

Sol. Answer (3)

$$\text{As } \log_{12} 180 + \log_{12} 25920 = \log_{12} ((180) \times (25920)) = 2 \log_{12} (2160)$$

Hence the numbers,

$$\log_{12} 180, \log_{12} 2160 \text{ and } \log_{12} 25920 \text{ are in A.P.}$$

$$\Rightarrow \log_{180} 12, \log_{2160} 12, \log_{25920} 12 \text{ are in H.P.}$$

97. The values of x such that $\log_{1/2} x > \log_{1/3} x$, is/are

(1) $x \in (0, 1)$

(2) $x \in [0, 1)$

(3) $x \in (0, 1]$

(4) $x \in [0, 1]$

Sol. Answer (1)

Using change of base formula;

$$\log_{1/3} x = \frac{\log_{1/2} x}{\log_{1/2} 1/3} = \log_{1/3} \left(\frac{1}{2} \right) \cdot \log_{1/2} x$$

Now, the inequality can be considered as;

$$\text{Now, } \log_{1/2} x \underbrace{\left\{ 1 - \log_{1/3} \frac{1}{2} \right\}}_{\oplus \text{ve}} > 0$$

$$\Rightarrow \log_{1/2} x > 0 \Rightarrow x < 1 \quad \dots(1)$$

$$\text{But original inequality is meaningful when } x > 0 \quad \dots(2)$$

$$\Rightarrow x \in (0, 1)$$

98. If the sum of first n terms of an A.P. is cn^2 , then the sum of squares of these n terms is **[IIT-JEE 2009]**

(1) $\frac{n(4n^2 - 1)c^2}{6}$

(2) $\frac{n(4n^2 + 1)c^2}{3}$

(3) $\frac{n(4n^2 - 1)c^2}{3}$

(4) $\frac{n(4n^2 + 1)c^2}{6}$

Sol. Answer (3)

$$S_n = cn^2$$

$$S_{n-1} = c(n-1)^2$$

$$t_n = c[n^2 - (n-1)^2] = c(2n-1)$$

$$\text{So now } \sum_{n=1}^n t_n^2 = c^2[1^2 + 3^2 + \dots + (2n-1)^2]$$

$$= c^2[1^2 + 2^2 + \dots + (2n)^2 - (2^2 + 4^2 + \dots + (2n)^2)]$$

$$= c^2 \left[\frac{(2n)(2n+1)(4n+1)}{6} - 2^2(1+2^2+\dots+n^2) \right]$$

$$= c^2 \left[\frac{(2n)(2n+1)(4n+1)}{6} - 4 \frac{(n)(n+1)(2n+1)}{6} \right]$$

$$= c^2 \left[\frac{(2n+1)}{3} (n(4n+1) - 2n(n+1)) \right]$$

$$= c^2 \left[\frac{(2n+1)}{3} (4n^2 + n - 2n^2 - 2n) \right]$$

$$\begin{aligned}
 &= c^2 \frac{(2n+1)}{3} (2n^2 - n) \\
 &= \frac{c^2 n(2n+1)(2n-1)}{3} \\
 &= \frac{c^2 n(4n^2 - 1)}{3}
 \end{aligned}$$

99. Let $a_1, a_2, a_3, \dots$ be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is [IIT-JEE 2012]

(1) 22 (2) 23 (3) 24 (4) 25

Sol. Answer (4)

$$\text{As, } \frac{1}{a_{20}} = \frac{1}{25} = \frac{1}{5} + 19d$$

$$\Rightarrow 19d = \left(\frac{1}{25} - \frac{1}{5} \right)$$

$$\Rightarrow d = -\frac{4}{25 \times 19}$$

$$\text{Also, } \frac{1}{a_n} = \frac{1}{5} - \frac{4}{25 \times 19} (n-1) < 0$$

$$\Rightarrow \frac{4(n-1)}{25 \times 19} > \frac{1}{5} \Rightarrow n > \frac{99}{4}$$

$\therefore$ The least integral value of n is 25

100. Let $b_i > 1$ for $i = 1, 2, \dots, 101$. Suppose $\log_e b_1, \log_e b_2, \dots, \log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, \dots, a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If $t = b_1 + b_2 + \dots + b_{51}$ and $s = a_1 + a_2 + \dots + a_{51}$, then [JEE(Advanced)-2016]

- (1) $s > t$ and $a_{101} > b_{101}$ (2) $s > t$ and $a_{101} < b_{101}$
 (3) $s < t$ and $a_{101} > b_{101}$ (4) $s < t$ and $a_{101} < b_{101}$

Sol. Answer (2)

$b_1, b_2, b_3, \dots, b_{101}$ are in G.P. with common ratio = 2

Let $a_1 = b_1 = a$

$$\text{Given, } a_{51} = b_{51} \Rightarrow a + 50d = a \cdot 2^{50}$$

$$t = a(2^{51} - 1)$$

$$s = \frac{51}{2} [a_1 + a_{51}] = \frac{51}{2} a(1 + 2^{50})$$

So, $s > t$

$$\text{Also, } a_{101} = a + 100d \text{ and } b_{101} = a \cdot 2^{100}$$

$$\frac{a_{101}}{b_{101}} = \frac{a + 2(a \cdot 2^{50} - a)}{a \cdot 2^{100}} = \frac{2^{51} - 1}{2^{100}} < 1$$

So, $a_{101} < b_{101}$

SECTION - B

Objective Type Questions (More than one options are correct)

1. If the Arithmetic mean of two positive numbers a and b ($a > b$) is twice their geometric mean, then $a : b$ equals

(1) $2 + \sqrt{3} : 2 - \sqrt{3}$

(2) $7 + 4\sqrt{3} : 1$

(3) $1 : 7 - 4\sqrt{3}$

(4) $2 : \sqrt{3}$

Sol. Answer (1, 2, 3)

If A.M. = A, G.M. = G

$$\text{Then } \frac{a}{b} = \frac{A + \sqrt{A^2 - G^2}}{A - \sqrt{A^2 - G^2}} \text{ or } \frac{A - \sqrt{A^2 - G^2}}{A + \sqrt{A^2 - G^2}}$$

But $A = 2G$

$$\Rightarrow \frac{a}{b} = \frac{2 + \sqrt{3}}{2 - \sqrt{3}} \text{ or } \frac{2 - \sqrt{3}}{2 + \sqrt{3}}$$

$$\text{also if } \frac{a}{b} = \frac{2 + \sqrt{3}}{2 - \sqrt{3}}$$

$$\text{then } \frac{a}{b} = \frac{2 + \sqrt{3}}{2 - \sqrt{3}} \times \frac{(2 + \sqrt{3})}{(2 + \sqrt{3})} = \frac{7 + 4\sqrt{3}}{1} = \frac{1}{7 - 4\sqrt{3}}$$

2. If the first and the $(2n + 1)^{\text{th}}$ terms of an A.P., a G.P. and an H.P. of positive terms are equal and their $(n + 1)^{\text{th}}$ terms are a , b and c respectively, then

(1) $a = b = c$

(2) $a + c = 2b$

(3) $a \geq b \geq c$

(4) $ac = b^2$

Sol. Answer (3, 4)

If total term = $(2n + 1)$ then middle term = $(n + 1)^{\text{th}}$

Let first term of all progressions is A and last term is B .

then

$$\text{For A.P., } a = \frac{A + B}{2}$$

$$\text{For G.P., } b = \sqrt{AB}$$

$$\text{For H.P., } c = \frac{2AB}{A + B}$$

$\Rightarrow a, b, c$ are A.M., G.M. and H.M. of A and B respectively.

Hence, $ac = b^2$ and $a \geq b \geq c$

3. If $\cos(\theta - \alpha)$, $\cos\theta$, $\cos(\theta + \alpha)$ are in H.P. and $\cos\alpha \neq 1$, then the angle α cannot lie in the

(1) I quadrant

(2) II quadrant

(3) III quadrant

(4) IV quadrant

Sol. Answer (1, 4)

$$\cos \theta = \frac{2\cos(\theta - \alpha)\cos(\theta + \alpha)}{\cos(\theta - \alpha) + \cos(\theta + \alpha)}$$

$$\cos \theta = \frac{\cos 2\theta + \cos 2\alpha}{2\cos \theta \cos \alpha}$$

$$\Rightarrow 2 \cos^2 \theta \cos \alpha = \cos 2\theta + \cos 2\alpha$$

$$\Rightarrow 2 \cos^2 \theta (1 - \cos \alpha) = 2 \sin^2 \alpha$$

$$\Rightarrow \sin^2 \frac{\alpha}{2} \left(\cos^2 \theta - 2 \cos^2 \frac{\alpha}{2} \right) = 0$$

$$\Rightarrow \cos^2 \frac{\alpha}{2} = \frac{1}{2} \cos^2 \theta$$

$$\cos \frac{\alpha}{2} = \pm \frac{1}{\sqrt{2}} \cos \theta \Rightarrow -\frac{1}{\sqrt{2}} \leq \cos \frac{\alpha}{2} \leq \frac{1}{\sqrt{2}} \Rightarrow \frac{\pi}{4} \leq \frac{\alpha}{2} \leq \frac{3\pi}{4}$$

$$\frac{\pi}{2} \leq \alpha \leq \frac{3\pi}{2}$$

Hence α does not lie in I and IV quadrant.

4. If x, y, z are in A.P. and x^2, y^2, z^2 are in H.P., then which of the following is correct?

- (1) x, y, z are all equal (2) $-\frac{x}{2}, y, z$ are in G.P. (3) $\frac{x}{2}, y, z$ are in G.P. (4) Either (1) or (2)

Sol. Answer (4)

$$2y = x + z$$

$$y^2 = \frac{2x^2 z^2}{x^2 + z^2}$$

By (i) and (ii)

$$\frac{(x+z)^2}{4} = \frac{2x^2 z^2}{(x^2 + z^2)}$$

$$\Rightarrow (x^2 + z^2 + 2xz)(x^2 + z^2) = 8x^2 z^2$$

$$(x^2 + z^2)^2 + 2xz(x^2 + z^2) - 8x^2 z^2 = 0$$

$$\Rightarrow (x^2 + z^2)^2 + 4xz(x^2 + z^2) - 2xz(x^2 + z^2) - 8x^2 z^2 = 0$$

$$\Rightarrow (x^2 + z^2)(x^2 + z^2 + 4xz) - 2xz(x^2 + z^2 + 4xz) = 0$$

$$\Rightarrow (x^2 + z^2 - 2xz)(x^2 + z^2 + 4xz) = 0$$

$$\Rightarrow (x-z)^2((x+z)^2 + 2xz) = 0$$

$$\Rightarrow x = z \text{ or } (x+z)^2 = -2xz$$

$$\Rightarrow 4y^2 = -2xz$$

$$\Rightarrow x = z \text{ or } y^2 = -\frac{xz}{2}$$

Hence $x = y = z$ or $-\frac{x}{2}, y, z$ are in G.P.

5. If a, b, c are in H.P., then

$$(1) \frac{a}{b+c-a}, \frac{b}{c+a-b}, \frac{c}{a+b-c} \text{ are in H.P.}$$

$$(2) \frac{2}{b} = \frac{1}{b-a} + \frac{1}{b-c}$$

$$(3) a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2} \text{ are in G.P.}$$

$$(4) \frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b} \text{ are in H.P.}$$

Sol. Answer (1, 2, 3, 4)

If a, b, c are in H.P.

Then $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

$$(1) \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ are in A.P.}$$

$$\Rightarrow \frac{a+b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c} \text{ are in A.P.}$$

$$\Rightarrow \frac{a+b+c}{a} - 2, \frac{a+b+c}{b} - 2, \frac{a+b+c}{c} - 2 \text{ are in A.P.}$$

$$\text{or } \frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c} \text{ are in A.P.}$$

$$\text{or } \frac{a}{b+c-a}, \frac{b}{c+a-b}, \frac{c}{a+b-c} \text{ are in H.P.}$$

Hence (1) is correct

$$(2) \frac{2}{b} = \frac{1}{b-a} + \frac{1}{b-c}$$

$$\frac{2}{b} = \frac{2b-a-c}{(b-a)(b-c)}$$

$$2(b^2 - bc - ab + ac) = 2b^2 - ab - bc$$

$$\Rightarrow 2b^2 - 2bc - 2ab + 2ac = 2b^2 - ab - bc$$

$$\Rightarrow -bc - ab = 2ac$$

$$b = \frac{2ac}{a+c}, \text{ hence } a, b, c \text{ are in H.P.}$$

$$(3) \left(\frac{b}{2}\right)^2 = \left(a - \frac{b}{2}\right)\left(c - \frac{b}{2}\right)$$

$$\frac{b^2}{4} = ac - \frac{ab}{2} - \frac{bc}{2} + \frac{b^2}{4}$$

$$\Rightarrow b = \frac{2ac}{a+c} \text{ hence } a, b, c \text{ are in H.P.}$$

$$(4) \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ are in A.P.}$$

$$\Rightarrow \frac{a+b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c} \text{ are in A.P.}$$

$$\Rightarrow \frac{a+b+c}{a} - 1, \frac{a+b+c}{b} - 1, \frac{a+b+c}{c} - 1, \text{ are in A.P.}$$

$$\Rightarrow \frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c} \text{ are in A.P.}$$

$$\Rightarrow \frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}, \text{ are in H.P.}$$

6. The sum of the first ten terms of an A.P. equals 155 and the sum of the first two terms of a G.P. equals 9. The first term of the A.P. is equal to the common ratio of the G.P. and the common difference of the A.P. is equal to the first term of G.P.. Given that the common difference of the A.P. is less than unity, which of the following is correct?

$$(1) \frac{25}{2} \text{ is a term of the G.P.}$$

$$(2) \frac{25}{3} \text{ is a term of the G.P.}$$

$$(3) \frac{25}{3} \text{ is a term of the A.P.}$$

$$(4) \frac{25}{2} \text{ is a term of the A.P.}$$

Sol. Answer (2, 4)

Let A. P. is

$$a, a + d, a + 2d, \dots$$

$$S_{10} = \frac{10}{2} [2a + 9d] = 155$$

$$\Rightarrow 2a + 9d = 31 \quad \dots(i)$$

Let G.P. is

$$A, AR, AR^2, \dots, AR^{n-1} \dots$$

$$A + AR = 9 \quad \dots(ii)$$

$$a = R \quad \dots(iii)$$

$$A = d \quad \dots(iv)$$

and it is given then $d < 1$

$$A + Aa = 9 \quad \dots(v)$$

By (i) and (ii), (iii), (iv)

$$A + A\left(\frac{31-9A}{2}\right) = 9$$

$$\Rightarrow 9A^2 - 33A + 18 = 0$$

$$9A^2 - 27A - 6A + 18 = 0$$

$$3A(3A - 9) - 2(3A - 9) = 0$$

$$\Rightarrow A = 3, A = \frac{2}{3}$$

if $A = 3$, then $d = 3$, $R = 2 \Rightarrow$ not possible as $d < 1$.

$$\text{if } A = \frac{2}{3}, \text{ then } d = \frac{2}{3}, R = \frac{25}{2}$$

Terms of G.P. = A, AR, AR^2

$$= \frac{2}{3}, \frac{25}{3}, \frac{625}{6}$$

Terms of A.P. = $a, a + d, a + 2d$

$$= \frac{25}{2}, \frac{25}{2} + \frac{2}{3}, \frac{25}{2} + \frac{4}{3}$$

Hence $\frac{25}{3}$ is a term of G.P.

and $\frac{25}{2}$ is a term of A.P.

7. In an equilateral triangle of side 3 cm, a circle is inscribed in which again an equilateral triangle is inscribed and so on. This continues for an infinite number of times. Then

- (1) Sum of areas of all circles (in cm^2) is $\sqrt{3}\pi$
- (2) Sum of areas of all circles (in cm^2) is π
- (3) Sum of areas of all equilateral triangles (in cm^2) is $3\sqrt{3}$
- (4) Sum of areas of all equilateral triangles (in cm^2) is $\sqrt{3}$

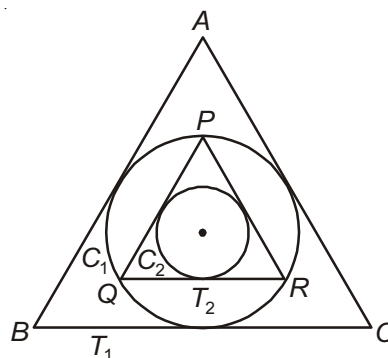
Sol. Answer (2, 3)

$$\text{The area of } \triangle ABC = A_1 = \frac{\sqrt{3}}{4} \times (3)^2$$

r_1 = The radius of circle C_1

= inradius of $\triangle ABC$

$$= \frac{\Delta}{S} = \frac{\frac{\sqrt{3}}{4} \times 3 \times 3}{\left(\frac{3+3+3}{2}\right)} = \frac{\frac{\sqrt{3}}{4} \times 9}{\left(\frac{9}{2}\right)} = \frac{\sqrt{3}}{2}$$



Let the area of circle C_1 is B_1 where $B_1 = \pi \left(\frac{\sqrt{3}}{2} \right)^2$

Again, let the side of equilateral triangle $\Delta PQR = a_1$

Circle C_1 is the circumcircle of ΔPQR .

$$r_1 = \frac{a_1}{2 \sin 60^\circ}$$

$$\frac{\sqrt{3}}{2} = \frac{a_1}{2 \times \frac{\sqrt{3}}{2}} \Rightarrow a_1 = \frac{3}{2}$$

$$\text{Area of equilateral triangle } \Delta PQR = A_2 = \frac{\sqrt{3}}{4} \times \left(\frac{3}{2} \right)^2$$

Again the radius of C_2 is the in radius of ΔPQR , Let that is r_2 .

$$r_2 = \frac{\Delta}{S} = \frac{\frac{\sqrt{3}}{4} \times \left(\frac{3}{2} \right)^2}{\frac{3}{2} \times \frac{3}{2}} = \frac{\frac{\sqrt{3}}{4} \times \frac{9}{4}}{\frac{9}{4}}$$

$$r_2 = \frac{\sqrt{3}}{4}$$

and so on

Sum of areas of equilateral triangles = $A_1 + A_2 + A_3 + \dots$

$$= \frac{\sqrt{3}}{4} \left(3^2 + \left(\frac{3}{2} \right)^2 + \left(\frac{3}{4} \right)^2 + \dots \text{upto } \infty \right)$$

$$= \frac{\sqrt{3}}{4} \times 3^2 \left(1 + \frac{1}{4} + \frac{1}{16} + \dots \right)$$

$$= \frac{\sqrt{3}}{4} \times 9 \times \frac{1}{1 - \frac{1}{4}} = \frac{\sqrt{3}}{4} \times 9 \times \frac{4}{3} = 3\sqrt{3} \text{ cm}^2$$

Sum of areas of all circles = $B_1 + B_2 + \dots$ upto ∞

$$= \pi r_1^2 + \pi r_2^2 + \dots$$

$$= \pi \left(\left(\frac{\sqrt{3}}{2} \right)^2 + \left(\frac{\sqrt{3}}{4} \right)^2 + \dots \right)$$

$$= 3\pi \left[\frac{1}{4} + \frac{1}{16} + \dots \right]$$

$$= 3\pi \left[\frac{\frac{1}{4}}{1 - \frac{1}{4}} \right] = 3\pi \times \frac{1}{3} = \pi \text{ cm}^2$$

8. Let $N = \frac{10^{143} - 1}{9}$. Then which of the following is true?

- (1) N is prime
- (2) $N = (1+10^{12}+10^{24}+\dots+10^{132})(1+10+10^2+\dots+10^{10})$
- (3) $N = (1+10^{11}+10^{22}+\dots+10^{132})(1+10+10^2+\dots+10^{10})$
- (4) $N = (1+10^{13}+10^{26}+\dots+10^{130})(1+10+10^2+\dots+10^{12})$

Sol. Answer (3, 4)

(1) N is not a prime because it is divisible by 9.

(2) $(1 + 10 + 10^2 + 10^3 + \dots + 10^{132})(1 + 10 + 10^2 + \dots + 10^{10})$

$$= \left(\frac{10^{133} - 1}{10^{12} - 1} \right) \left(\frac{10^{11} - 1}{10^2 - 1} \right) \neq \left(\frac{10^{143} - 1}{9} \right)$$

Hence (2) is false.

(3) $(1 + 10^{11} + 10^{22} + 10^{33} + \dots + 10^{132})(1 + 10 + 10^2 + \dots + 10^{10})$

$$= \left(\frac{(10^{11})^{13} - 1}{10^{11} - 1} \right) \left(\frac{10^{11} - 1}{10 - 1} \right) = \frac{10^{143} - 1}{10^{11} - 1} \times \frac{10^{11} - 1}{10 - 1}$$

$$= \frac{10^{143} - 1}{9} = N$$

Hence option (3) is true.

(4) $(1 + 10^{13} + 10^{26} + \dots + 10^{130})(1 + 10 + 10^2 + \dots + 10^{12})$

$$= \left(\frac{(10^{13})^{11} - 1}{10^{13} - 1} \right) \times \left(\frac{10^{13} - 1}{10 - 1} \right) = \frac{10^{143} - 1}{9} = N$$

Hence (4) is true

9. If $\log_x a$, $a^{x/2}$ and $\log_b x$ are in GP, then x is equal to

- (1) $\log_a(\log_b a)$
- (2) $\log_a(\log_e a) - \log_a(\log_e b)$
- (3) $-\log_a(\log_a b)$
- (4) $\log_a(\log_e b) - \log_a(\log_e a)$

Sol. Answer (1, 2)

$\log_x a$, $a^{x/2}$, $\log_b x$ are in GP

$$\therefore \left(a^{x/2} \right)^2 = \log_x a \cdot \log_b x$$

$$\Rightarrow a^x = \log_b a$$

$$\Rightarrow x = \log_a(\log_b a)$$

$$\text{Now, } \log_a(\log_e a) - \log_a(\log_e b)$$

10. The sum of the squares of three distinct real numbers which are in GP is S^2 , if their sum is αS , then

(1) $1 < \alpha^2 < 3$

(2) $\frac{1}{3} < \alpha^2 < 1$

(3) $1 < \alpha < 3$

(4) $\frac{1}{3} < \alpha < 1$

Sol. Answer (1, 2)

Let the numbers are, $\frac{a}{r}, a, ar$

Now, we have,

$$\frac{a^2}{r^2} + a^2 + a^2 r^2 = S^2 \text{ and } \frac{a}{r} + a + ar = \alpha S$$

Let, $r + \frac{1}{r} = x$, $x \neq \pm 2$ as $r \neq \pm 1$

$$a^2 \left[\left(r + \frac{1}{r} \right)^2 - 2 \right] + a^2 = S^2$$

$$\Rightarrow a^2 [x^2 - 2] + a^2 = S^2$$

$$\Rightarrow a^2 [x^2 - 1] = S^2 \quad \dots (I)$$

$$ax + a = \alpha S$$

$$\Rightarrow a^2 (x + 1)^2 = \alpha^2 S^2 \quad \dots (II)$$

Dividing (I) by (II) we get,

$$\frac{x^2 - 1}{(x + 1)^2} = \frac{1}{\alpha^2}$$

$$\Rightarrow \frac{x - 1}{x + 1} = \frac{1}{\alpha^2}$$

$$\Rightarrow \frac{x - 1 + x + 1}{x - 1 - x - 1} = \frac{1 + \alpha^2}{1 - \alpha^2} \Rightarrow x = \frac{1 + \alpha^2}{\alpha^2 - 1}$$

$$\Rightarrow r + \frac{1}{r} = \frac{\alpha^2 + 1}{\alpha^2 - 1}$$

$$\Rightarrow (r^2 + 1)(\alpha^2 - 1) = r(\alpha^2 + 1)$$

$$\Rightarrow (\alpha^2 - 1)r^2 - r(\alpha^2 + 1) + (\alpha^2 - 1) = 0$$

$$\therefore D \geq 0$$

$$\Rightarrow (\alpha^2 + 1)^2 - 4(\alpha^2 - 1)^2 \geq 0$$

$$\Rightarrow [\alpha^2 + 1 + 2(\alpha^2 - 1)][\alpha^2 + 1 - 2(\alpha^2 - 1)] \geq 0$$

$$\Rightarrow (3\alpha^2 - 1)(-\alpha^2 + 3)$$

$$\frac{1}{3} \leq \alpha^2 \leq 3$$

$$\text{But } \alpha^2 \neq 1, \frac{1}{3}, 3$$

$$\frac{1}{3} < \alpha^2 < 1, 1 < \alpha^2 < 3$$

11. The equation $(\log_x 10)^3 - (\log_x 10)^2 - 6 \log_x 10 = 0$ is satisfied by a value of x given by

(1) $\sqrt[3]{10}$

(2) $\frac{1}{\sqrt{10}}$

(3) $\sqrt{10}$

(4) $\frac{1}{\sqrt[3]{10}}$

Sol. Answer (1, 2)

$$\text{Let } \log_x 10 = y$$

$$y^3 - y^2 - 6y = 0$$

$$y(y^2 - y - 6) = 0 \Rightarrow y(y - 3)(y + 2) = 0$$

$$\Rightarrow y = 0, y = 3, y = -2$$

$$\Rightarrow \log_x 10 = 0, 3, -2$$

$$\log_{10} x = \infty, \frac{1}{3}, -\frac{1}{2} \text{ (but } \infty \text{ is not defined)}$$

$$\Rightarrow \log_{10} x = \frac{1}{3}, \frac{1}{2}$$

$$x = 10^{\frac{1}{3}}, 10^{-\frac{1}{2}}$$

12. Let $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$. Then S_n can take value(s)

[JEE(Advanced) 2013]

(1) 1056

(2) 1088

(3) 1120

(4) 1332

Sol. Answer (1, 4)

$$S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$$

$$S_n = -1^2 - 2^2 + 3^2 + 4^2 - 5^2 - 6^2 + \dots - (4n-3)^2 - (4n-2)^2 + (4n-1)^2 + (4n)^2$$

$$S_n = (3^2 - 1^2) + (4^2 - 2^2) + (7^2 - 5^2) + (8^2 - 6^2) + (11^2 - 9^2) + (12^2 - 10^2) + \dots + (4n-1)^2 - (4n-3)^2 + (4n)^2 - (4n-2)^2$$

$$S_n = 2(1+3) + 2(4+2) + 2(7+5) + 2(8+6) + \dots + 2(4n-1+4n-3) + 2(4n+4n-2)$$

$$S_n = 2[1+2+3+\dots+4n] = \frac{2 \cdot 4n(4n+1)}{2}$$

From A, $4n(4n + 1) = 1056$

$$4n^2 + n = 264$$

$$4n^2 + n - 264 = 0$$

$$n = 8$$

From B, $4n(4n + 1) = 1088$ (Not possible)

From C, $4n(4n + 1) = 1120$ (Not possible)

From D, $4n(4n + 1) = 1332$

$$n = 9$$

13. If $I = \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)} dx$, then

[JEE(Advanced) 2017]

(1) $I > \frac{49}{50}$

(2) $I < \frac{49}{50}$

(3) $I < \log_e 99$

(4) $I > \log_e 99$

Sol. Answer (1, 3)

$$\begin{aligned} I &= \sum_{k=1}^{98} (k+1) \left(\int_k^{k+1} \frac{dx}{x(x+1)} \right) \\ &= \sum_{k=1}^{98} (k+1) \left\{ \log \left(\frac{x}{1+x} \right) \right\}_k^{k+1} = \sum_{k=1}^{98} (k+1) \left\{ \log \left(\frac{k+1}{k+2} \right) - \log \left(\frac{k}{k+1} \right) \right\} \\ &= \sum_{k=1}^{98} \left\{ (k+1) \log \left(\frac{k+1}{k+2} \right) - k \log \left(\frac{k}{k+1} \right) \right\} + \{ \log(k+1) - \log k \} \\ &= \left\{ 99 \log \left(\frac{99}{100} \right) - \log \left(\frac{1}{2} \right) \right\} + (\log 99 - \log 1) \\ &= 99 \log \left(\frac{99}{100} \right) + \log 2 + \log_e (99) \end{aligned}$$

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

The ubiquitous AM–GM inequality has many applications. It almost crops up in unlikely situations and the solutions using AM–GM are truly elegant. Recall that for n positive reals a_i , $i = 1, 2, \dots, n$, the AM–GM inequality tells

$$\frac{\sum_{i=1}^n a_i}{n} \geq \left(\prod_{i=1}^n a_i \right)^{\frac{1}{n}}$$

The special case in which the inequality turns into equality helps solve many problems where at first we seem to have not enough information to arrive at the answer.

1. If the equation $x^4 - 4x^3 + ax^2 + bx + 1 = 0$ has four positive roots, then the value of $\frac{|a| + |b|}{a + b}$ is
- (1) 5 (2) -5 (3) 3 (4) -3

Sol. Answer (1)

Let the roots are x_1, x_2, x_3, x_4 and all are positive

By the properties of roots

$$x_1 + x_2 + x_3 + x_4 = 4 \quad \dots(i)$$

$$x_1x_2 + x_1x_3 + x_1x_4 + x_2x_3 + x_2x_4 + x_3x_4 = a \quad \dots(ii)$$

$$x_1x_2x_3 + x_1x_2x_4 + x_2x_3x_4 + x_3x_4x_1 = -b \quad \dots(iii)$$

$$x_1x_2x_3x_4 = 1 \quad \dots(iv)$$

By (i) and (iv)

A.M. of $x_1, x_2, x_3, x_4 =$ G.M. of x_1, x_2, x_3, x_4

Hence the roots are equal

$$\Rightarrow x_1 = x_2 = x_3 = x_4 = 1 \quad \text{by (i)}$$

$$\Rightarrow a = 6, \text{ and } b = -4$$

$$\frac{|a| + |b|}{a + b} = \frac{6 + 4}{6 - 4} = \frac{10}{2} = 5$$

2. The number of ordered pairs (x, y) of real numbers satisfying the equation $x^8 + 6 = 8|xy| - y^8$ is
- (1) 10 (2) 8 (3) 4 (4) 2

Sol. Answer (3)

$$x^8 + y^8 + 6 = 8xy$$

Here we use A.M. $\geq$ G.M. for x^8, y^8 and 6.

$$\Rightarrow \frac{x^8 + y^8 + 1 + 1 + 1 + 1 + 1 + 1}{8} \geq (x^8 \cdot y^8 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 1)^{1/8}$$

$$\Rightarrow x^8 + y^8 + 6 \geq 8|xy| \quad \dots (ii)$$

By (i) and (ii) A.M. = G.M. $\Rightarrow$ number are equal

$$\text{Hence, } x^8 = 1 \Rightarrow x = \pm 1$$

$$y^8 = 1 \Rightarrow y = \pm 1$$

Hence total 4 order pairs $(1, 1), (1, -1), (-1, 1), (-1, -1)$ are formed.

3. If a, b, c are positive integers satisfying

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = \frac{3}{2}, \text{ then the value of } abc + \frac{1}{abc}$$

(1) Is $\frac{85}{4}$

(2) Is $\frac{17}{4}$

(3) Is $\frac{65}{8}$

(4) Can't be determined

Sol. Answer (3)

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = \frac{3}{2}$$

$$\Rightarrow \left(\frac{a}{b+c} + 1 \right) + \left(\frac{b}{c+a} + 1 \right) + \left(\frac{c}{a+b} + 1 \right) = \frac{3}{2} + 3 \Rightarrow \frac{a+b+c}{b+c} + \frac{a+b+c}{c+a} + \frac{a+b+c}{a+b} = \frac{9}{2}$$

$$\Rightarrow (a+b+c) \left(\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a} \right) = \frac{9}{2} \quad \dots(i)$$

Now let us take three numbers $a+b, b+c, c+a$ and using A.M $\geq$ H.M.

$$\frac{(a+b) + (b+c) + (c+a)}{3} \geq \frac{3}{\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a}}$$

$$\Rightarrow \frac{2(a+b+c)}{3} \geq \frac{3}{\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a}}$$

$$\Rightarrow (a+b+c) \left(\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a} \right) \geq \frac{9}{2} \quad \dots(ii)$$

By (i) and (ii)

We find that

$$A.M = H.M.$$

Hence numbers are equal this is possible only

$$\text{when } a = b = c$$

$$\text{Let } a = b = c = 2$$

$$\text{Then } abc + \frac{1}{abc} = 8 + \frac{1}{8} = \frac{65}{8}$$

Hence only (3) is correct

Comprehension-II

One useful way of defining sequences is by a recursion relation. Many recurrence relations can be transformed to some known sequences, say GP or sometimes n th term can be found by algebraic jugglery.

1. If $a_n - a_{n-1} = 1$ for every positive integer greater than 1, then $a_1 + a_2 + a_3 + \dots + a_{100}$ equals

(1) $5000 \cdot a_1$ (2) $5050 \cdot a_1$ (3) $5051 \cdot a_1$ (4) $5052 \cdot a_1$

Sol. Answer (2)

$$a_n - a_{n-1} = 1$$

$$\Rightarrow a_2 = 2, a_3 = 3, \dots, a_n = n$$

$$a_1 + a_2 + \dots + a_{100} = 1 + 2 + 3 + \dots + 100 = 5050$$

2. Some chocolates are distributed between 25 children in such a way that first child gets 5 chocolates, second child gets 7 chocolates and in general, n th child gets n chocolates more than the $(n-1)$ th child. The total number of chocolates distributed is

(1) 3250 (2) 2525 (3) 2750 (4) 3025

Sol. Answer (4)

$$S = 5 + 7 + 10 + 14 + \dots + T_n \quad \dots (i)$$

$$\text{Again } S = 5 + 7 + 10 + \dots + T_{n-1} + T_n \quad \dots (ii)$$

By subtracting (ii) from (i)

$$0 = 5 + 2 + 3 + 4 + \dots + (T_n - T_{n-1}) - T_n$$

$$T_n = 5 + 2 + 3 + 4 + \dots + (T_n - T_{n-1})$$

$$T_n = 5 + [2 + 3 + 4 + \dots + (n-1) \text{ terms}]$$

$$= 5 + \left(\frac{n-1}{2} \right) (n+2) = \frac{n^2 + n + 8}{2}$$

$$S = \sum T_n = \frac{1}{2} (\sum n^2 + \sum n + 8 \sum 1) = \frac{1}{2} \left(\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + 8n \right)$$

At $n = 25$

$$S = \frac{1}{2} \left[\frac{25 \times 26 \times 51}{6} + \frac{25 \times 26}{2} + 8 \times 25 \right] = 3025$$

3. Let $\{a_n\}$ be a sequence defined by $a_1 = 1$

$$a_1 + 2a_2 + 3a_3 + \dots + (n-1)a_{n-1} = n^2 a_n, \quad n \geq 2$$

The value of a_{786} is

(1) $\frac{2}{789}$ (2) $\frac{1}{393}$ (3) $\frac{2}{393}$ (4) $\frac{1}{1572}$

Sol. Answer (4)

$$a_1 = 1$$

$$\text{and } a_1 + 2a_2 + 3a_3 + \dots (n-1)a_{n-1} = n^2 a_n \quad \dots(i)$$

using (i)

$$\text{at } n = 2, \quad a_1 = 4a_2 \Rightarrow a_2 = \frac{1}{4}$$

$$\text{at } n = 3, \quad a_1 + 2a_2 = 9a_3$$

$$\Rightarrow 1 + 2 \times \frac{1}{4} = 9a_3 \quad \Rightarrow 1 + \frac{1}{2} = 9a_3 \Rightarrow a_3 = \frac{1}{6}$$

$$\text{Similarly } a_4 = \frac{1}{8}, \text{ and so on.}$$

$$\Rightarrow a_n = \frac{1}{2n}$$

$$a_{786} = \frac{1}{2 \times 786} = \frac{1}{1572}$$

Comprehension-III

If A , G and H are respectively arithmetic, geometric and harmonic means between a and b both being unequal and positive, then

$$A = \frac{a+b}{2} \Rightarrow a+b = 2A$$

$$G = \sqrt{ab} \Rightarrow G^2 = ab$$

$$H = \frac{2ab}{a+b} \Rightarrow G^2 = AH$$

On the basis of above information answer the following questions.

1. If the geometric and harmonic means of two numbers are 16 and $12\frac{4}{5}$, then the ratio of one number to the other is

(1) 1 : 4

(2) 2 : 3

(3) 1 : 2

(4) 2 : 1

Sol. Answer (1)

$$\text{Since, } \sqrt{ab} = 16 \text{ and } \frac{2ab}{a+b} = \frac{64}{5}$$

$$\therefore \frac{2 \times 256}{a+b} = \frac{64}{5}$$

$$\Rightarrow a+b = \frac{2 \times 256 \times 5}{64}$$

$$\Rightarrow a+b = 40 \Rightarrow a = 8, b = 32$$

$$\therefore a : b = 1 : 4$$

2. The sum of AM and GM of two positive numbers equal to the difference between the numbers. The numbers are in the ratio

(1) $1 : 3$

(2) $1 : 6$

(3) $9 : 1$

(4) $1 : 12$

Sol. Answer (3)

$$\text{Since, } \frac{a+b}{2} + \sqrt{ab} = a - b$$

$$\Rightarrow 2\sqrt{ab} = a - 3b$$

$$\Rightarrow 4ab = a^2 - 6ab + 9b^2$$

$$\Rightarrow a^2 - 10ab + 9b^2 = 0$$

$$\Rightarrow a^2 - 9ab - ab + 9b^2 = 0$$

$$\Rightarrow a(a - 9b) - b(a - 9b) = 0$$

$$\Rightarrow (a - b)(a - 9b) = 0$$

$$\Rightarrow a = 9b \Rightarrow a : b = 9 : 1$$

$$\Rightarrow (\text{as } a \neq b)$$

3. The numbers whose A.M. and G.M. are A and G is

(1) $A \pm (A^2 - G^2)$

(2) $\sqrt{A} \pm \sqrt{A^2 - G^2}$

(3) $A \pm \sqrt{A^2 - G^2}$

(4) $\frac{A \pm \sqrt{A^2 - G^2}}{2}$

Sol. Answer (3)

Let two numbers be a and b

$$\text{AM} = \frac{a+b}{2} = A$$

$$\text{and, GM} = \sqrt{ab} = G$$

$$\text{Now, } (a-b)^2 = (a+b)^2 - 4ab$$

$$= 4A^2 - 4G^2$$

$$= 4(A^2 - G^2)$$

$$\Rightarrow a - b = \pm 2\sqrt{A^2 - G^2}$$

Now,

$$a + b = 2A$$

$$a - b = \pm 2\sqrt{A^2 - G^2}$$

Solving, we get

$$a = A \pm \sqrt{A^2 - G^2}$$

$$\text{and } b = A \pm \sqrt{A^2 - G^2}$$

Comprehension-IV

While solving logarithmic equations or logarithmic inequalities care must be taken to ensure that the value of the variable obtained do indeed satisfy the given equation. Often the solution consists in transforming the original equation to a form which can be solved with ease. But in bargain the process of solution brings with it extraneous solutions, because the transformations carried out are not always equivalent. In what follows one must verify that the values of variables obtained indeed satisfy original equation or inequation.

1. How many solutions in real numbers does the equation $3^x 8^{\frac{x}{x+2}} = 6$ have?

- (1) None (2) Exactly one (3) Exactly two (4) Infinite

Sol. Answer (3)

$$3^x 8^{\frac{x}{x+2}} = 6 \Rightarrow 3^x 2^{\frac{3x}{x+2}} = 2.3$$

$$\Rightarrow 2^{\frac{3x}{x+2}-1} = 3^{1-x}$$

Taking log on both sides with base 2,

$$\Rightarrow \frac{3x}{x+2} - 1 = (1-x)\log_2 3$$

$$\Rightarrow \frac{2(x-1)}{x+2} = (1-x)\log_2 3$$

$$\Rightarrow x-1=0, \frac{2}{x+2} = \log_2 3$$

$$\Rightarrow x=1, x \neq 2, \frac{2}{x+2} = \frac{1}{\log_3 2}$$

$$x = 2\log_3 2 - 2$$

$$x = 2[\log_3 2 - 1] = 2[\log_3 2 - \log_3 3]$$

$$x = 2\log_3 \frac{2}{3} = \log_3 \left(\frac{4}{9}\right)$$

$$\Rightarrow x=1, x = \log_3 2^{-2} - 2$$

$$\Rightarrow x=1, x = \log_3 \frac{1}{36}$$

$\therefore$ Equation has exactly two solutions.

2. The solution of the inequality

$$x \log_{\frac{1}{10}} (x^2 + x + 1) > 0$$

is given by

- (1) $-\infty < x < 2$ (2) $-\infty < x < -1$ (3) $-3 < x < 1$ (4) $1 < x < 3$

Sol. Answer (2)

$$x \log_{\left(\frac{1}{10}\right)} (x^2 + x + 1) > 0$$

$$\Rightarrow x \log_{10} (x^2 + x + 1) < 0$$

$$\text{here } x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} > 0, \text{ for all } x \in \mathbb{R}$$

Now again

$$x \log_{10} (x^2 + x + 1) < 0$$

here two cases arise

$$(i) \quad x > 0 \quad \dots(i)$$

$$\text{and } \log_{10} (x^2 + x + 1) < 0$$

$$\Rightarrow x^2 + x + 1 < 1 \Rightarrow x^2 + x < 0 \Rightarrow x(x + 1) < 0$$

$$\text{or } x \in (-1, 0) \quad \dots(ii)$$

by (i) and (ii), we find that no value of x exists

$$(ii) \quad x < 0, \text{ and } \log_{10} (x^2 + x + 1) > 0$$

$$x < 0, \quad x^2 + x + 1 > 10^0$$

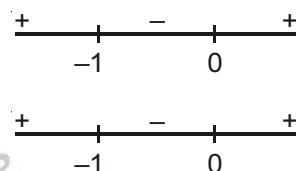
$$x < 0 \quad \text{and} \quad x^2 + x + 1 > 1$$

$$\Rightarrow x < 0 \text{ and } x^2 + x + 1 > 1 \Rightarrow x^2 + x > 0$$

$$x < 0 \text{ and } x \in (-\infty, -1) \cup (0, \infty)$$

$$\Rightarrow x \in (-\infty, -1)$$

Hence option (2) is correct.



3. Let S be the set of all solutions x in real numbers of the equation $(\log_5 x)^2 + \log_{5x} \frac{5}{x} = 1$. Then $\sum_{x \in S} x$ equals

- (1) 126 (2) $\frac{151}{25}$ (3) $\frac{131}{25}$ (4) 131

Sol. Answer (2)

We have,

$$(\log_5 x)^2 + \log_{5x} \left(\frac{5}{x}\right) = 1$$

$$\Rightarrow (\log_5 x)^2 + \log_{5x} 5 - \log_{5x} x = 1$$

$$(\log_5 x)^2 + \frac{1}{\log_5 5x} - \frac{1}{\log_x 5x} = 1$$

$$(\log_5 x)^2 + \frac{1}{(1 + \log_5 x)} - \frac{1}{(\log_x 5) + 1} = 1$$

Let $\log_5 x = y$

$$y^2 + \frac{1}{1+y} - \frac{1}{\frac{1}{y} + 1} = 1$$

$$y^2 + \frac{1}{1+y} - \frac{y}{y+1} = 1$$

$$\Rightarrow y^2(1+y) + 1 - y = (y+1)$$

$$y^2 + y^3 + 1 - y = y + 1$$

$$y^3 - 2y + y^2 = 0$$

$$y^3 + y^2 - 2y = 0$$

$$y(y^2 + y - 2) = 0$$

$$y(y+2)(y-1) = 0$$

$$y = 0, y = -2, y = 1$$

$$\Rightarrow \log_5 x = 0, -2, 1$$

$$\Rightarrow x = 5^0, 5^{-2}, 5$$

$$\Rightarrow x = 1, \frac{1}{25}, 5$$

$$\text{Sum of value of } x = 1 + \frac{1}{25} + 5 = 6 + \frac{1}{25} = \frac{151}{25}$$

Comprehension-V

Let V_r denote the sum of the first r terms of an arithmetic progression (A.P.) whose first term is r and the common difference is $(2r-1)$. Let $T_r = V_{r+1} - V_r - 2$ and $Q_r = T_{r+1} - T_r$ for $r = 1, 2, \dots$ [IIT-JEE 2007]

1. The sum $V_1 + V_2 + \dots + V_n$ is

$$(1) \frac{1}{12}n(n+1)(3n^2 - n + 1)$$

$$(2) \frac{1}{12}n(n+1)(3n^2 + n + 2)$$

$$(3) \frac{1}{2}n(2n^2 - n + 1)$$

$$(4) \frac{1}{3}(2n^3 - 2n + 3)$$

2. T_r is always

(1) An odd number

(2) An even number

(3) A prime number

(4) A composite number

3. Which one of the following is a correct statement?

(1) $Q_1, Q_2, Q_3, \dots$ are in A.P. with common difference 5

(2) $Q_1, Q_2, Q_3, \dots$ are in A.P. with common difference 6

(3) $Q_1, Q_2, Q_3, \dots$ are in A.P. with common difference 11

(4) $Q_1 = Q_2 = Q_3 = \dots$

Solution of Comprehension-V

1. Answer (2)
2. Answer (4)
3. Answer (2)

V_r turns out be $\frac{r}{2}(2r^2 - r + 1)$

$$V_1 = 1$$

$$V_2 = 7$$

wrong choices can be eliminated using these two values.

$T_r = (r + 1)(3r - 1)$, obviously a composite number $Q_r = T_{r+1} - T_r = 6r + 5$, which is an ΔP with common difference 6.

Comprehension-VI

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n \geq 2$, let A_{n-1} and H_{n-1} have arithmetic, geometric and harmonic means as A_n, G_n, H_n respectively. **[IIT-JEE 2007]**

1. Which one of the following statements is correct?

(1) $G_1 > G_2 > G_3 > \dots$

(2) $G_1 < G_2 < G_3 < \dots$

(3) $G_1 = G_2 = G_3 = \dots$

(4) $G_1 < G_3 < G_5 < \dots$ and $G_2 > G_4 > G_6 > \dots$

Sol. Answer (3)

2. Which one of the following statements is correct?

(1) $A_1 > A_2 > A_3 > \dots$

(2) $A_1 < A_2 < A_3 < \dots$

(3) $A_1 > A_3 > A_5 > \dots$ and $A_2 < A_4 < A_6 < \dots$

(4) $A_1 < A_3 < A_5 < \dots$ and $A_2 > A_4 > A_6 > \dots$

Sol. Answer (1)

3. Which one of the following statements is correct?

(1) $H_1 > H_2 > H_3 > \dots$

(2) $H_1 < H_2 < H_3 < \dots$

(3) $H_1 > H_3 > H_5 > \dots$ and $H_2 < H_4 < H_6 < \dots$

(4) $H_1 < H_3 < H_5 < \dots$ and $H_2 > H_4 > H_6 > \dots$

Sol. Answer (2)

SECTION - D**Assertion-Reason Type Questions**

1. STATEMENT-1 : If a, b, c are distinct positive reals in G.P., then $\log_a n, \log_b n, \log_c n$ are in H.P., $\forall n \in \mathbb{N}$
and

STATEMENT-2 : The reciprocals of terms of Arithmetic progression with non-zero terms, form an H.P.

Sol. Answer (4)

Statement 1 is false for $n = 1$

But statement 2 is clearly true.

2. STATEMENT-1 : The sum of reciprocals of first n terms of the series $1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{9} + \dots$ is n^2 .

and

STATEMENT-2 : A sequence is said to be H.P. if the reciprocals of its terms are in A.P.

Sol. Answer (2)

Sum of reciprocals

$= 1 + \frac{1}{3} + \frac{1}{5} + \dots$ upto n terms

$$= \frac{n}{2} [1 \times 1 + (n-1) \times 2] = n^2$$

Hence statement 1 is true and the statement 2 is clearly true.

3. STATEMENT-1 : For three positive unequal quantities a, b, c are in H.P., we must have $a^{2008} + c^{2008} > 2b^{2008}$.

and

STATEMENT-2 : A.M. $\geq$ G.M. $\geq$ H.M. for positive numbers.

Sol. Answer (1)

Using A.M $\geq$ G.M.

$$\frac{a^{2008} + c^{2008}}{2} \geq (\sqrt{ac})^{2008}$$

But $\sqrt{ac} > b$

$$\Rightarrow a^{2008} + c^{2008} \geq 2(\sqrt{ac})^{2008} > 2b^{2008}$$

Hence statement 1 is clearly true statement (2) is also true and explains statement 1.

4. STATEMENT-1 : The sum of 100 arithmetic means between two given numbers 1000 and 3016 is 200800.

and

STATEMENT-2 : The sum of n arithmetic means between two given numbers is n^{th} power of their single A.M.

Sol. Answer (3)

$$\text{Sum} = \frac{100}{2} [1000 + 3016] = \frac{401600}{2} = 200800$$

hence statement-1 is true

statement (2) is false because

sum $= n(\text{single A.M.})$

5. STATEMENT-1 : If a, b, c, d are positive and distinct numbers in H.P., then $a + d > b + c$.

and

STATEMENT-2 : If a, b, c, d are in H.P., then $\frac{a+d}{ad} = \frac{b+c}{bc}$.

Sol. Answer (2)

$$\frac{a+c}{2} > b, \frac{b+d}{2} > c \Rightarrow a+d > b+c [\because \text{AM} > \text{HM}]$$

$$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d} \text{ in AP} \Rightarrow \frac{1}{b} - \frac{1}{a} = \frac{1}{d} - \frac{1}{c} \Rightarrow \frac{1}{a} + \frac{1}{d} = \frac{1}{b} + \frac{1}{c}$$

$$\Rightarrow \frac{a+d}{ad} = \frac{b+c}{bc}$$

$\therefore$ Statement 1 and statement 2 both are true but statement 2 is not correct reason for statement 1

6. STATEMENT-1 : For $n \in N, n > 1, n+1 > 2(n!)^{\frac{1}{n}}$.

and

STATEMENT-2 : A.M. of distinct positive number is greater than G.M.

Sol. Answer (1)

Statement 2 is obviously true

$$\frac{1+2+3+\dots+n}{n} > (1.2.3. \dots n)^{\frac{1}{n}} [\text{AM} > \text{GM}]$$

$$\Rightarrow \frac{n(n+1)}{2n} > (n!)^{\frac{1}{n}}$$

$$\Rightarrow n+1 > 2(n!)^{\frac{1}{n}}$$

7. STATEMENT-1 : The sum of n terms of two arithmetic progressions are in A.P. in the ratio $(7n+1) : (4n+17)$, then the ratio of their n^{th} terms is $7 : 4$.

and

STATEMENT-2 : If $S_n = ax^2 + bx + c$, then $T_n = S_n - S_{n-1}$.

Sol. Answer (4)

$$\frac{S_n}{S_n'} = \frac{7n+1}{4n+17} = \frac{n(7n+1)}{n(4n+17)}$$

$$\Rightarrow S_n = (7n^2 + n)\lambda \quad S_n' = (4n^2 + 17n)$$

$$\frac{T_n}{T_n'} = \frac{S_n - S_{n-1}}{S_n' - S_{n-1}'} = \frac{7(2n-1)+1}{4(2n-1)+17} = \frac{14n-6}{8n+13}$$

8. STATEMENT-1 : If the angle of a convex polygon are in A.P. $120^\circ, 125^\circ, 130^\circ, \dots$, then it has 16 sides.

and

STATEMENT-2 : The sum of the angles of a polygon of x sides is $(n-2) 180^\circ$.

Sol. Answer (4)

Let n be the number of sides

$$\frac{n}{2}[240 + (n-1)5^\circ] = (n-2)180^\circ \Rightarrow n^2 - 25n + 144 = 0$$

$$\therefore n = 9, 16 \Rightarrow \text{Largest angle } 120^\circ + 15 \times 5^\circ = 195^\circ > 180^\circ$$

Which is not possible in a convex polygon.

9. STATEMENT-1 : If the infinite A.G.P. $1, \sqrt{3}, 2, x, \dots$ has a finite sum, then $x = 2$.

and

STATEMENT-2 : The infinite A.G.P. $a, (a+d)r, (a+2d)r^2, \dots$ has a finite sum only if $|r| < 1$.

Sol. Answer (1)

$$\sqrt{3} = (1+d)r, \quad 2 = (1+2d)r^2$$

$$\Rightarrow 1 = \frac{2\sqrt{3}}{r} - \frac{2}{r^2}$$

$$\Rightarrow r^2 - 2\sqrt{3}r + 2 = 0 \Rightarrow r = \sqrt{3} \pm 1 \text{ Taking}$$

$$r = \sqrt{3} - 1, \quad d = \frac{\sqrt{3}}{r} - 1 \quad \therefore d = \frac{\sqrt{3}}{\sqrt{3}-1} - 1 = \frac{\sqrt{3}+1}{2}$$

$$\begin{aligned} x &= (1+3d)r^3 = \left[1 + \frac{3}{2}(\sqrt{3}+1)\right](\sqrt{3}-1)^3 = \left(\frac{5+3\sqrt{3}}{2}\right)(\sqrt{3}-1)(4-2\sqrt{3}) \\ &= (5+3\sqrt{3})(\sqrt{3}-1)(2-\sqrt{3}) = (3\sqrt{3}+5)(3\sqrt{3}-5) = 27-25 = 2 \end{aligned}$$

10. STATEMENT-1 : If $a^x = b^y = c^z$; where x, y, z are unequal positive numbers and a, b, c are in G.P., then $x^3 + z^3 > 2y^3$.

and

STATEMENT-2 : If a, b, c are in H.P. $a^3 + c^3 \geq 2b^3$, where a, b, c are positive real numbers.

Sol. Answer (1)

$$\text{Given } a^x = b^y = c^z = k \Rightarrow a = k^{1/x}, \quad b = k^{1/y}, \quad c = k^{1/z}$$

$$\therefore a, b, c \text{ are in GP} \Rightarrow \frac{b}{a} = \frac{c}{b} \Rightarrow k^{1/y-1/x} = k^{1/z-1/y}$$

$$\Rightarrow \frac{2}{y} = \frac{1}{x} + \frac{1}{z} \text{ Hence } x, y, z \text{ are in H.P.}$$

$$\therefore y \text{ is H.M. of } x \text{ and } z \Rightarrow \sqrt{xz} > y$$

$$\frac{x^3 + z^3}{2} > (\sqrt{xz})^3 > y^3 \Rightarrow x^3 + z^3 > 2y^3$$

11. Let $\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b}$

STATEMENT-1 : $a^a b^b c^c = 1$.

and

STATEMENT-2 : $a^{b+c} \cdot b^{c+a} \cdot c^{a+b} = 1$.

Sol. Answer (2)

$$\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b} = K \text{ (Let)}$$

$$\Rightarrow \log a = K(b-c)$$

$$\log b = K(c-a)$$

$$\log c = K(a-b)$$

Also, $a, b, c > 0$

Statement-1 :

Let $y = a^a b^b c^c$

$$\log y = a \log a + b \log b + c \log c$$

$$= aK(b-c) + b \cdot K(c-a) + c \cdot K(a-b) = 0$$

$$\Rightarrow y = 1, \text{ hence statement 1 is true.}$$

Statement-2 :

Let $y = a^{b+c} \cdot b^{c+a} \cdot c^{a+b}$

$$\log y = (b+c) \log a + (c+a) \log b + (a+b) \log c$$

$$= (b+c) \cdot K(b-c) + (c+a) K(c-a) + (a+b) K(a-b)$$

$$= K[b^2 - c^2 + c^2 - a^2 + a^2 - b^2] = 0$$

$$\Rightarrow y = 1$$

Hence statement-2 is also true and explains statement-1.

12. STATEMENT-1 : $\log_4 16$ is a rational number.

and

STATEMENT-2 : $\log_2 3$ is an irrational number.

Sol. Answer (2)

$\log_4 16 = \log_4 4^2 = 2 \log_4 4 = 2$, a rational number but $\log_2 3$ is an irrational number.

Because we may prove this by following method.

Let $\log_2 3$ is a rational number $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

$$\Rightarrow \log_2 3 = \frac{p}{q} \Rightarrow 3 = \frac{2^p}{2^q} \Rightarrow 3^q = 2^p, \text{ which is not possible for integers } p \text{ and } q, \text{ hence } \log_2 3 \text{ is not a rational number. } \log_2 3 \text{ is an irrational number.}$$

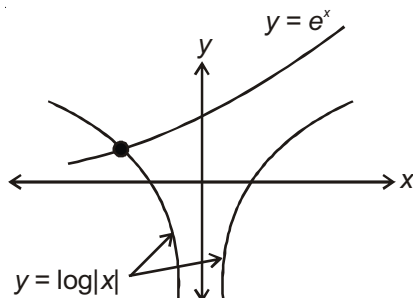
13. STATEMENT-1: Number of solution of $\log|x| = e^x$ is two.

and

STATEMENT-2: If $\log_{30}3 = a$, $\log_{30}5 = b$ then $\log_{30}8 = 3(1 - a - b)$.

Sol. Answer (4)

$e^x = \log|x|$ has only one solution



14. Suppose four distinct positive numbers a_1, a_2, a_3, a_4 are in G.P. Let $b_1 = a_1, b_2 = b_1 + a_2, b_3 = b_2 + a_3$ and $b_4 = b_3 + a_4$. **[IIT-JEE 2008]**

STATEMENT-1 : The numbers b_1, b_2, b_3, b_4 are neither in A.P. nor in G.P.

and

STATEMENT-2 : The numbers b_1, b_2, b_3, b_4 are in H.P.

Sol. Answer (3)

Let $a_1 = 1, a_2 = 2, a_3 = 4, a_4 = 8$

$\therefore b_1 = 1, b_2 = 3, b_3 = 7, b_4 = 15$

Clearly, b_1, b_2, b_3, b_4 are not in HP.

Statement-2 is false.

SECTION - E

Matrix-Match Type Questions

1. Match the following

Column-I

(A) If $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in H.P., then

(B) If a, b, c are in A.P., then

(C) If a, b, c are in G.P., then

(D) If sum of roots of the quadratic $ax^2 + bx + c = 0$ ($abc \neq 0$) equals to the sum of squares of their reciprocals then

Column-II

(p) $\log_a x, \log_b x, \log_c x$ are in H.P.

(Assume that log functions are defined)

(q) $a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ are in A.P.

(r) $\frac{c}{a}, \frac{a}{b}, \frac{b}{c}$ are in A.P.

(s) $\frac{ab+ac}{bc}, \frac{bc+ba}{ca}, \frac{ca+cb}{ab}$ are in A.P.

Sol. Answer A(q, s), B(q, s), C(p), D(r)

(A) If $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in H.P.

$\Rightarrow a, b, c$ are in A.P.

Subtract $\frac{b}{2}$ from each term

$a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ are in A.P.

(B) a, b, c are in A.P.

Divide each term by abc

$\Rightarrow \frac{1}{bc}, \frac{1}{ca}, \frac{1}{ab}$ are in A.P.

multiply each term by $(ab + bc + ca)$

$\Rightarrow \frac{ab + bc + ca}{bc}, \frac{ab + bc + ca}{ca}, \frac{ab + bc + ca}{ab}$ are in A.P.

$\Rightarrow \frac{ab + ac}{bc} + 1, \frac{bc + ab}{ca} + 1, \frac{bc + ca}{ab} + 1$ are in A.P.

$\Rightarrow \frac{ab + ac}{bc}, \frac{bc + ab}{ca}, \frac{ca + ca}{ab}$ are in A.P.

(C) If a, b, c are in G.P.

then $\log a, \log b, \log c$ are in A.P.

or $\frac{1}{\log a}, \frac{1}{\log b}, \frac{1}{\log c}$ are in H.P.

$\Rightarrow \frac{\log x}{\log a}, \frac{\log x}{\log b}, \frac{\log x}{\log c}$ are in H.P.

$\Rightarrow \log_a x, \log_b x, \log_c x$ are in A.P.

(D) $\alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$, where α, β are the roots

$$\Rightarrow \alpha + \beta = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2}$$

$$(\alpha + \beta) = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$\left(-\frac{b}{a}\right) = \frac{\frac{b^2}{a^2} - 2\frac{c}{a}}{\left(\frac{-c}{a}\right)^2}$$

$$\left(-\frac{b}{a}\right) = \frac{\frac{b^2 - 2ac}{a^2}}{\frac{c^2}{a^2}} = \frac{b^2 - 2ac}{c^2}$$

$$-bc^2 = b^2a - 2a^2c$$

$$\Rightarrow 2a^2c = b^2a + bc^2$$

$$\Rightarrow 2\frac{a}{b} = \frac{b}{c} + \frac{c}{a}$$

$$\Rightarrow \frac{c}{a}, \frac{a}{b}, \frac{b}{c} \text{ are in A.P.}$$

2. Match the equations with their solutions

Column-I (Equations)

Column-II (Solutions)

(A) $x \left[\frac{1}{1.2.3.4} + \frac{1}{2.3.4.5} + \dots \text{to } \infty \right] = 2$

(p) 72

(B) $x \left[\frac{1}{1.3} + \frac{2}{1.3.5} + \frac{3}{1.3.5.7} + \dots \text{to } \infty \right] = 20$

(q) 80

(C) $x \left[\frac{3}{1.2.4} + \frac{4}{2.3.5} + \frac{5}{3.4.6} + \dots \text{to } \infty \right] = 58$

(r) 36

(D) $x \left[\left(\frac{2}{1.3} \right) \frac{1}{3} + \left(\frac{3}{3.5} \right) \frac{1}{3^2} + \left(\frac{4}{5.7} \right) \frac{1}{3^3} + \dots \text{to } \infty \right] = 20$

(s) 40

Sol. Answer A(r), B(s), C(p), D(q)

(A) $x \left[\frac{1}{1.2.3.4} + \frac{1}{2.3.4.5} + \dots \right] = 2$

$$\Rightarrow \frac{x}{3} \left[\left(\frac{1}{1.2.3} - \frac{1}{2.3.4} \right) + \left(\frac{1}{2.3.4} - \frac{1}{3.4.5} \right) + \dots \right] = 2$$

$$\Rightarrow \frac{x}{3} \left[\frac{1}{1.2.3} \right] = 2$$

$$\Rightarrow x = 18 \times 2 = 36$$

(B) $x \left[\frac{1}{1.3} + \frac{2}{1.3.5} + \frac{3}{1.3.5.7} + \dots \right] = 20$

$$\frac{x}{2} \left[\frac{2}{1.3} + \frac{4}{1.3.5} + \frac{6}{1.3.5.7} + \dots \right] = 20$$

$$\frac{x}{2} \left[\left(1 - \frac{1}{1.3} \right) + \left(\frac{1}{1.3} - \frac{1}{1.3.5} \right) + \left(\frac{1}{1.3.5} - \frac{1}{1.3.5.7} \right) + \dots \right] = 20$$

$$\Rightarrow \frac{x}{2} (1) = 20$$

$$\Rightarrow x = 40$$

$$\begin{aligned}
 \text{(C)} \quad T_n &= \frac{(n+2)}{n(n+1)(n+3)} = \frac{(n+2)^2}{n(n+1)(n+2)(n+3)} \\
 &= \frac{n(n+1)+3n+4}{n(n+1)(n+2)(n+3)} \\
 &= \frac{1}{(n+2)(n+3)} + \frac{3}{(n+1)(n+2)(n+3)} + \frac{4}{n(n+1)(n+2)(n+3)}
 \end{aligned}$$

$$\text{Hence, } S_n = K - \frac{1}{n+3} - \frac{3}{2(n+2)(n+3)} - \frac{4}{3(n+1)(n+2)(n+3)} \quad [\text{Using the properties of series}]$$

$$\text{At } n = 1, k = \frac{29}{36}$$

$$S_n = \frac{29}{36} - \frac{1}{n+3} - \frac{3}{2(n+2)(n+3)} - \frac{4}{3(n+1)(n+2)(n+3)}$$

$$S_\infty = \frac{29}{36}$$

$$x \left[\frac{29}{36} \right] = 58$$

$$\Rightarrow x = 72$$

$$\begin{aligned}
 \text{(D)} \quad T_n &= \left(\frac{n+1}{(2n-1)(2n+1)} \right) \cdot \frac{1}{3^n} \\
 &= \left(\frac{3}{4} \cdot \frac{1}{(2n-1)} - \frac{1}{4(2n+1)} \right) \cdot \frac{1}{3^n} \\
 &= \frac{1}{4} \left[\frac{3}{2n-1} - \frac{1}{2n+1} \right] \cdot \frac{1}{3^n} = \frac{1}{4} \left[\frac{1}{2n-1} \cdot \frac{1}{3^{n-1}} - \frac{1}{(2n+1)} \cdot \frac{1}{3^n} \right]
 \end{aligned}$$

$$S_n = \frac{1}{4} [1 - 0] = \frac{1}{4}$$

$$\Rightarrow \frac{x}{4} = 20 \Rightarrow x = 80$$

3. Match the series with their sum

Column-I

Column-II

- | | |
|--|-----------|
| (A) $1.2.3.4 + 2.3.4.5 + \dots + n(n+1)(n+2)(n+3) + \dots$ to 11 terms | (p) 10385 |
| (B) $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \dots + \frac{1^3+2^3+\dots+n^3}{1+3+\dots+(2n-1)} + \dots$ to 16 terms | (q) 37820 |
| (C) $1.3 + 2.4 + \dots + n(n+2) + \dots$ to 30 terms | (r) 446 |
| (D) $1 + 3 + \dots + \frac{n(n+1)}{2} + \dots$ to 60 terms | (s) 72072 |

Sol. Answer A(s), B(r), C(p), D(q)

$$(A) T_n = n(n+1)(n+2)(n+3)$$

$$\Rightarrow S_n = \frac{n(n+1)(n+2)(n+3)(n+4)}{5} + C$$

$$\text{At } n = 1$$

$$1.2.3.4 = \frac{1 \times 2 \times 3 \times 4 \times 5}{5} + C$$

$$\Rightarrow C = 0$$

$$\Rightarrow S_{11} = \frac{11 \times 12 \times 13 \times 14 \times 15}{5} = 72072$$

$$(B) T_n = \frac{1^3 + 2^3 + \dots + n^3}{1 + 3 + 5 + \dots + (2n-1)}$$

$$= \frac{\left(\frac{n(n+1)}{2}\right)^2}{n^2} = \frac{1}{4}(n+1)^2$$

$$T_n = \frac{1}{4}(n+1)^2$$

$$S_{16} = \frac{1}{4}[2^2 + 3^2 + 4^2 + \dots + 17^2]$$

$$= \frac{1}{4}[(1^2 + 2^2 + \dots + 17^2) - 1]$$

$$= \frac{1}{4}\left[\frac{17 \times 18 \times 35}{6} - 1\right]$$

$$= \frac{1}{4}[1785 - 1] = \frac{1784}{4} = 446$$

$$(C) T_n = n(n+2) = n^2 + 2n$$

$$S_n = \sum n^2 + \sum 2n$$

$$S_n = \frac{n(n+1)(2n+1)}{6} + n(n+1)$$

$$S_{30} = \frac{30 \times 31 \times 61}{6} + 30 \times 31$$

$$= 5 \times 31 \times 61 + 30 \times 31$$

$$= 10385$$

$$(D) T_n = \frac{n(n+1)}{2} = \frac{1}{2}(n^2 + n)$$

$$S_n = \frac{1}{2}[\sum n^2 + \sum n]$$

$$S_n = \frac{1}{2}\left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}\right]$$

$$= \frac{n(n+1)(n+2)}{6}$$

$$\Rightarrow S_{60} = \frac{60 \times 61 \times 62}{6}$$

$$= 37820$$

4. Match the following entries of column I with those of column II.

Column I

Column II

- (A) If the roots of the equation $x^3 - 9x^2 + 23x - 15 = 0$

are in AP, then their common difference is

(p) ${}^2C_1 \cdot {}^2P_0$

- (B) Number of values of x for which $[x]$, $\operatorname{sgn} x$, $\{x\}$ $\{x \neq 0\}$

are in AP, is

(q) 5

- (C) If a , b , c are in AP, then the straight line

$ax + by + c = 0$ will always passed through a fixed point whose ordinate and abscissae are

(r) 1

- (D) An infinite G.P. has the first term x and sum 5, then

x may be

(s) 2

(t) - 2

Sol. Answer A(p, s, t), B(p, s), C(r, t), D(p, q, r, s)

- (A) The roots of the given equation are 1, 3, 5

Hence, common difference is $= \pm 2$

- (B) Obviously, for $x = \pm 2$, $[x]$, $\operatorname{sgn} x$ and $\{x\}$

Will be in A.P.

Hence there are two values of x .

- (C) Since, a , b , c are in A.P.

$\Rightarrow 2b = a + c$, then straight line $ax + by + c = 0$ will pass through $(1, -2)$ because it satisfy the condition $a - 2b + c = 0$ or $2b = a + c$

(D) We have, sum of infinite G.P

$$S_{\infty} = \begin{cases} \frac{a}{1-r}, & |r| < 1 \\ \infty, & |r| \geq 1 \end{cases}$$

$$\therefore S_{\infty} = \frac{x}{1-r} = S$$

$$\Rightarrow 1-r = \frac{x}{5}$$

$$\Rightarrow r = \frac{5-x}{5} \text{ exists only when } |r| < 1$$

$$\Rightarrow -1 < \frac{5-x}{5} < 1$$

$$\Rightarrow -5 < 5-x < 5$$

$$\Rightarrow -10 < -x < 0$$

$$\Rightarrow 0 < x < 10$$

5. Match the following entries of column I with those of column II

Column I

Column II

(A) Let, $S_1, S_2, \dots$ be squares such that for each $n \geq 1$, the length of a side of S_n equals the length of a diagonal of S_{n+1} . If the length of a side of S_1 is 10 cm. The value of n for which area of S_n is less than 1 sq.cm is

(p) 2

(B) If the first two terms of a HP are $\frac{2}{5}$ and $\frac{12}{13}$ respectively. If the largest term is T , then $[T]$ is less than or equal to

(q) 4

(C) The value of $(0.16)^{\log_{2.5}\left(\frac{1}{3} + \frac{1}{3^2} + \dots \infty\right)}$ is less than or equal to

(r) 6

(D) If $3 + \frac{1}{4}(3+d) + \frac{1}{4^2}(3+2d) + \dots \infty$, then d is greater than or equal to

(s) 8

(t) 9

Sol. Answer A(s, t), B(p, q, r, s, t), C(q, r, s, t), D(p, q, r, s, t)

(A) We have,

$$\begin{aligned}\text{Length of side of } S_n &= \text{Length of diagonal of } S_{n+1} \\ &= \sqrt{2} (\text{Length of side of } S_{n+1})\end{aligned}$$

$$\Rightarrow \frac{\text{Length of side of } S_n}{\text{Length of side of } S_{n+1}} = \sqrt{2} \text{ for all } n \geq 1$$

$$\Rightarrow \text{Sides of } S_1, S_2, \dots, S_n \text{ form a GP with common ratio } \frac{1}{\sqrt{2}} \text{ and first term } 10$$

$$\therefore \text{Side of } S_n = 10 \cdot \left(\frac{1}{\sqrt{2}}\right)^{n-1} = \frac{10}{2^{\frac{n-1}{2}}}$$

$$\therefore \text{Area of } S_n = \frac{100}{2^{n-1}}$$

$$\text{Now, } S_n < 1$$

$$\therefore \frac{100}{2^{n-1}} < 1 \Rightarrow 2^{n-1} > 100$$

$$\therefore n \geq 8$$

$$(C) \text{ Let, } S = \frac{1}{3} + \frac{1}{3^2} + \dots \infty$$

$$= \frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{1}{2}$$

$$\text{Let } y = (.16)^{\log_{2.5} \frac{1}{2}}$$

$$\Rightarrow y = (0.4)^{2 \log} (0.4)^{-1 \left(\frac{1}{2}\right)}$$

$$\Rightarrow (0.4)^{\log} 0.4^{(4)} = 4$$

$$(D) \frac{3}{1 - \frac{1}{4}} + \frac{d \times \frac{1}{4}}{\left(1 - \frac{1}{4}\right)^2} = 8 \quad \left[\text{using } S = \frac{a}{1-r} + \frac{dr}{(1-r)^2}\right]$$

$$\Rightarrow d = 9$$

SECTION - F

Integer Answer Type Questions

1. If H.M. of two numbers is 4, then A.M. 'A' and G.M. 'G' satisfy the relation $2A + G^2 = 27$, then modulus of difference of these two numbers is

Sol. Answer (3)

Let that numbers be a, b

$$\text{Then } \frac{2ab}{a+b} = 4$$

$$A = \frac{a+b}{2}, G = \sqrt{ab}$$

$$2\left(\frac{a+b}{2}\right) + (\sqrt{ab})^2 = 27$$

$$\Rightarrow a + b + ab = 27$$

$$a + b + 2(a + b) = 27$$

$$a + b = 9, ab = 18$$

$$(a - b)^2 = (a + b)^2 - 9ab = 81 - 72$$

$$a - b = 9$$

$$a - b = \pm 3$$

When $a - b = 3$ when $a - b = -3$

$$a + b = 9, a = 3, b = 6$$

$$a = 6, b = 3$$

$$|a - b| = 3$$

2. Let $a_1, a_2, \dots, a_{10}$ be in AP and $n_1, n_2, \dots, n_{10}$ be in HP, if $a_1 = n_1 = 2$ and $a_{10} = n_{10} = 3$ then $a_4 a_7$ is ____

Sol. Answer (6)

Let d the common difference of the AP, then

$$a_{10} = 3 \Rightarrow a_1 + 9d = 3 \Rightarrow 2 + 9d = 3 \Rightarrow d = \frac{1}{9}$$

$$\therefore a_4 = a_1 + 3d = 2 + 3 \cdot \frac{1}{9} = 2 + \frac{1}{3} = \frac{7}{3}$$

Let D be the common difference of $\frac{1}{n_1}, \frac{1}{n_2}, \dots, \frac{1}{n_{10}}$

$$\text{Then } \frac{1}{n_{10}} = \frac{1}{n_1} + 9D$$

$$\Rightarrow \frac{1}{3} = \frac{1}{2} + 9D$$

$$\Rightarrow 9D = \frac{1}{3} - \frac{1}{2} \Rightarrow D = -\frac{1}{54}$$

$$\frac{1}{n_7} = \frac{1}{n_1} + 6D$$

$$= \frac{1}{2} + 6 \cdot -\frac{1}{54} = \frac{1}{2} - \frac{1}{9}$$

$$= \frac{9-2}{18} = \frac{7}{18}$$

$$\therefore a_4 \cdot a_7 = \frac{7}{3} \times \frac{18}{7} = 6$$

3. If a_n be the n^{th} term of an AP and if $a_7 = 15$, then the value of the common difference that would make $a_2 a_7 a_{12}$ greatest is _____.

Sol. Answer (0)

Let d be the common difference of the A.P. then,

$$a_2 a_7 a_{12} = (15 - 5d) 15 (15 + 5d) = 375 (9 - d^2)$$

The value of $a_2 a_7 a_{12}$ will be greatest when $d = 0$

4. If a, b, c, d are in GP, then the value of $(a - c)^2 + (b - c)^2 + (b - d)^2 - (a - d)^2$ is _____.

Sol. Answer (0)

Let r be the common ratio of G.P.

$$\therefore b = ar, c = ar^2, d = ar^3$$

$$\begin{aligned} \therefore (a - c)^2 + (b - c)^2 + (b - d)^2 - (a - d)^2 &= (a - ar^2)^2 + (ar - ar^2)^2 + (ar - ar^3)^2 - (a - ar^3)^2 \\ &= a^2(1 - r^2)^2 + a^2 r^2(1 - r)^2 + a^2 r^2(1 - r^2)^2 - a^2(1 - r^3)^2 \\ &= a^2(1 - r)^2 + [(1 + r)^2 + r^2 + r^2(1 + r)^2 - (1 + r + r^2)^2] \\ &= a^2(1 - r)^2 + [1 + r^2 + 2r + r^2 + r^2(1 + 2r + r^2) - (1 + r^2 + r^4 + 2r + 2r^3 + 2r^2)] \\ &= a^2(1 - r)^2 + [1 + r^2 + 2r + r^2 + r^2 + 2r^3 + r^4 - 1 - r^2 - r^4 - 2r - 2r^3 - 2r^2] \\ &= a^2(1 - r)^2 (0) = 0 \end{aligned}$$

5. If the m^{th} term of a HP be n and the n^{th} term be m , then the $(mn)^{\text{th}}$ term is _____.

Sol. Answer (1)

Let a be the first term and d be the common difference of the corresponding A.P., then

$$n = m^{\text{th}} \text{ term} = \frac{1}{a + (m-1)d} \Rightarrow a + (m-1)d = \frac{1}{n}$$

$$\text{Similarly } a + (n-1)d = \frac{1}{m}$$

$$\text{Solving above two equation we get, } a = \frac{1}{mn}, d = \frac{1}{mn}$$

So, $(mn)^{\text{th}}$ term of the H.P.

$$= \frac{1}{(mn)^{\text{th}} \text{ term of corresponding A.P.}}$$

$$= \frac{1}{\frac{1}{mn} + (mn-1) \cdot \frac{1}{mn}}$$

$$= \frac{1}{\frac{1}{mn} + 1 - \frac{1}{mn}} = \frac{1}{1} = 1$$

6. The number of solution of $x^{1+\log x} = 10x$ is _____.

Sol. Answer (2)

Taking log both sides.

$$(1 + \log x) \log x = (1 + \log x); x > 0$$

$$\Rightarrow (1 + \log x)(\log x - 1) = 0$$

$$\therefore \log x = 1, -1$$

$$\Rightarrow x = 10, \frac{1}{10}$$

Number of solution 2

7. Let S_k , $k = 1, 2, \dots, 100$, denote the sum of the infinite geometric series whose first term is $\frac{k-1}{k!}$ and the common ratio is $\frac{1}{k}$. Then the value of $\frac{100^2}{100!} + \sum_{k=1}^{100} |(k^2 - 3k + 1)S_k|$ is

[IIT-JEE 2010]

Sol. Answer (4)

We have,

$$S_k = \frac{\frac{k-1}{k!}}{1 - \frac{1}{k}} = \frac{1}{(k-1)!}$$

$$\text{Now, } (k^2 - 3k + 1)S_k = \{(k-2)(k-1) - 1\} \times \frac{1}{(k-1)!} = \frac{1}{(k-3)!} - \frac{1}{(k-1)!}$$

$$\Rightarrow \sum_{k=1}^{100} |(k^2 - 3k + 1)S_k| = 1 + 1 + 2 - \left(\frac{1}{99!} + \frac{1}{98!}\right) = 4 - \frac{100^2}{100!}$$

$$\Rightarrow \frac{100^2}{100!} + \sum_{k=1}^{100} |(k^2 - 3k + 1)S_k| = 4$$

8. Let $a_1, a_2, a_3, \dots, a_{11}$ be real numbers satisfying $a_1 = 15, 27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4, \dots, 11$. If $\frac{a_1^2 + a_2^2 + \dots + a_{11}^2}{11} = 90$, then the value of $\frac{a_1 + a_2 + \dots + a_{11}}{11}$ is equal to **[IIT-JEE 2010]**

Sol. Answer (0)

9. Let $a_1, a_2, a_3, \dots, a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^p a_i, 1 \leq p \leq 100$. For any integer n with $1 \leq n \leq 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n , then a_2 is **[IIT-JEE 2011]**

Sol. Answer (9)

We have,

$$a_1 = 3, S_p = a_1 + a_2 + a_3 + \dots + a_p, 1 \leq p \leq 100$$

By the given condition,

$$\frac{S_m}{S_n} = \frac{m}{n} \frac{6 + (m-1)d}{6 + (n-1)d} = k, \text{ which is independent of } m \text{ and } n$$

$$\Rightarrow 5(6 + (m-1)d) = k(6 + (n-1)d)$$

$$\Rightarrow 30 + 25nd - 5d = 6k + nkd - kd$$

which is independent of n

Hence, on comparing the coefficients of like terms, we get

$$k = 25$$

$$30 - 5d = 150 - 25d$$

$$\Rightarrow 20d = 120$$

$$\Rightarrow d = 6$$

$$a_2 = 9$$

10. The minimum value of the sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} with $a > 0$ is **[IIT-JEE 2011]**

Sol. Answer (8)

For $a > 0$,

$$\frac{a^{-5} + a^{-4} + 3a^{-3} + 1 + a^8 + a^{10}}{8} \geq \sqrt[8]{a^{-5} \cdot a^{-4} \cdot a^{-9} \cdot 1 \cdot a^8 \cdot a^{10}}$$

$$\Rightarrow a^{-5} + a^{-4} + 3a^{-3} + 1 + a^8 + a^{10} \geq 8$$

$$\Rightarrow \min(a^{-5} + a^{-4} + 3a^{-3} + 1 + a^8 + a^{10}) = 8$$

11. A pack contains n cards numbered from 1 to n . Two consecutive numbered cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller of the numbers on the removed cards is k , then $k - 20 =$ **[JEE(Advanced) 2013]**

Sol. Answer (5)

The smallest value of n for which

$$\frac{n(n+1)}{2} > 1224$$

$$n(n+1) > 2448 \Rightarrow n > 49$$

For $n = 50$

$$\frac{n(n+1)}{2} \Rightarrow 1275$$

$$\text{So, } k + (k+1) = 1275 - 1224 = 51$$

$$k = 25$$

$$k - 20 = 5$$

12. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is $b + 2$, then the value of $\frac{a^2 + a - 14}{a + 1}$ is **[JEE(Advanced) 2014]**

Sol. Answer (4)

$$a, ar, ar^2, 0 > 1 \quad r \text{ is integer}$$

$$\frac{a + b + c}{3} = b + 2$$

$$a + ar + ar^2 = 3(b + 2)$$

$$a + ar + ar^2 = 3ar + 6$$

$$ar^2 - 2ar + a - 6 = 0$$

$$a(r-1)^2 = 6$$

$$a = 6, r = 2$$

$$\text{So } \frac{a^2 + a - 14}{a + 1} = \frac{6^2 + 6 - 14}{6 + 1} = \frac{28}{7} = 4$$

13. Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is $6 : 11$ and the seventh term lies in between 130 and 140, then the common difference of this A.P. is **[JEE(Advanced) 2015]**

Sol. Answer (9)

$$\frac{\frac{7}{2}[2a + 6d]}{\frac{11}{2}[2a + 10d]} = \frac{6}{11}$$

$$\Rightarrow \frac{a + 3d}{a + 5d} = \frac{6}{7} \Rightarrow a = 9d$$

$$\text{Also, } 130 < a + 6d < 140$$

$$130 < 15d < 140$$

$$\Rightarrow d = 9$$

14. The sides of a right angled triangle are in arithmetic progression. If the triangle has area 24, then what is the length of its smallest side? **[JEE(Advanced) 2017]**

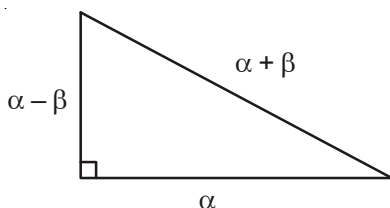
Sol. Answer (6)

Using $(\alpha + \beta)^2 = \alpha^2 + (\alpha - \beta)^2$

$$\Rightarrow \beta = \frac{\alpha}{4}$$

Also, $\frac{1}{2}\alpha(\alpha - \beta) = 24$

$$\Rightarrow \left. \begin{array}{l} \alpha = 8 \\ \beta = 2 \end{array} \right\}$$



Smallest side = 6

15. Let X be the set consisting of the first 2018 terms of the arithmetic progression 1, 6, 11, ..., and Y be the set consisting of the first 2018 terms of the arithmetic progression 9, 16, 23, Then, the number of elements in the set $X \cup Y$ is _____. [JEE(Advanced) 2018]

Sol. Answer (3748)

$$X = \{1, 6, 11, \dots, 10086\}$$

$$Y = \{9, 16, 23, \dots, 14128\}$$

First common term is 16 with common difference 35

$$t_n = 16 + (n - 1)35$$

$$= 35n - 19 \leq 10086$$

$$\Rightarrow n \leq 288.71$$

Hence number of common terms is 288

$$n(X \cup Y) = n(X) + n(Y) - n(X \cap Y) = 2018 + 2018 - 288 = 3748$$

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : If x is positive, then least value of $x^{12} + \frac{4}{x^2} + \frac{4}{x}$ is 9.

STATEMENT-2 : If $a^2 + b^2 + c^2 = 4$, $x^2 + y^2 + z^2 = 9$ then maximum value of $ax + by + cz$ is 6.

STATEMENT-3 : $2 + \frac{3}{2^2} + \frac{4}{2^3} + \dots$ upto $\infty = 4$.

(1) F T F

(2) T T F

(3) T T T

(4) F F T

Sol. Answer (3)

Statement-1 :

By A.M. $\geq$ G.M.

$$\begin{aligned} & \frac{x^{12} + \left(\frac{1}{x^2} + \frac{1}{x^2} + \frac{1}{x^2} + \frac{1}{x^2}\right) + \left(\frac{1}{x} + \frac{1}{x} + \frac{1}{x} + \frac{1}{x}\right)}{9} \geq \sqrt[9]{x^{12} \times x^{\frac{1}{8}} \times x^{\frac{1}{4}}} \\ \Rightarrow & \frac{x^{12} + \frac{4}{x^2} + \frac{4}{x}}{9} \geq \sqrt[9]{1} \\ \Rightarrow & x^{12} + \frac{4}{x^2} + \frac{4}{x} \geq 9 \end{aligned}$$

Statement-2 :

$$\text{As, } (a^2 + b^2 + c^2)(x^2 + y^2 + z^2)$$

$$(ax + by + cz)^2 \geq 0$$

$$\Rightarrow (ax + by + cz)^2 \leq 36$$

$$\Rightarrow \sqrt{ax + by + cz} \leq 6$$

Statement-3 :

$$S = \frac{3}{2^2} + \frac{4}{2^3} + \frac{5}{2^4} + \dots \infty$$

$$\frac{1}{2}S = \frac{3}{2^3} + \frac{4}{2^4} + \dots \infty$$

$$S - \frac{1}{2}S = \frac{3}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots \infty$$

$$S = 2$$

$$\text{Therefore } 2 + \frac{3}{2^2} + \frac{4}{2^3} + \dots \text{ upto } \infty = 2 + 2 = 4$$

Statement -3 is true

2. STATEMENT-1 : If a, b, c are in A.P., b, c, a are in G.P. then c, a, b are in H.P.STATEMENT-2 : If $a_1, a_2, a_3, \dots, a_{100}$ are in A.P. and $a_3 + a_{98} = 50$ then $a_1 + a_2 + a_3 + \dots + a_{100} = 2500$.STATEMENT-3 : If $a_1, a_2, \dots, a_n \in R$ then $\frac{a_1 + a_2 + a_3 + \dots + a_n}{n} \geq (a_1 \cdot a_2 \cdot \dots \cdot a_n)^{1/n}$.

(1) F T F

(2) T T F

(3) T T T

(4) F F T

Sol. Answer (2)**Statement-1:**

$$b = \frac{a+c}{2} \quad \dots(i)$$

$$c = \sqrt{ab} \quad \dots(ii)$$

We have to prove

$$a = \frac{2bc}{b+c} \text{ from (1) \& (2) we get}$$

 $a = c$ required result

Statement-1 is true.

Statement-2:

$$a_3 + a_{98} = 50$$

then, $a_1 + a_2 + a_3 + \dots + a_{100}$

$$= (a_1 + a_{100}) + (a_2 + a_{99}) + (a_3 + a_{98}) + \dots 50 \text{ terms}$$

$$= 50 \times 50 = 2500$$

Statement-2 is true.

Statement-3:

If $a_1, a_2, \dots, a_m > 0$

3. STATEMENT-1 : $1^2 - 2^2 + 3^2 \dots$ to 21 terms is 231

STATEMENT-2 : $1^3 - 2^3 + 3^3 - 4^3 \dots$ to 15 terms is 1856

STATEMENT-3 : $1^2 + 3^2 + 5^2 \dots$ to 8 terms is 689

(1) T T F

(2) T F F

(3) T F T

(4) T T T

Sol. Answer (1)

Statement-1:

$$(1^2 - 2^2) + (3^2 - 4^2) \dots (19^2 - 20^2) + 21^2$$

$$= (1 - 2)(1 + 2) + (3 - 4)(3 + 4) + \dots$$

$$= -(1 + 2 + 3 + \dots + 20) + 21^2 = 441 - 210 = 231$$

Statement-2 :

$$1^3 + 2^3 + 3^3 + \dots + 15^3 - 2(2^3 + 4^3 \dots + 14^3)$$

$$= \left(\frac{15 \times 16}{2}\right)^2 - 16(1^3 + 2^3 + \dots + 7^3) = 120^2 - 16\left(\frac{7 \times 8}{2}\right)^2$$

$$= 120^2 - 112^2 = 1856$$

Statement-3 :

$$1^2 + 2^2 + 3^2 \dots + 15^2 - (2^2 + 4^2 \dots + 14^2)$$

$$= \frac{15 \times 16 \times 31}{6} - 4(1^2 + 2^2 + \dots + 7^2) = 40 \times 31 - \frac{4 \cdot 7 \cdot 8 \cdot 15}{6}$$

$$= 40(31 - 14) = 40 \times 17 = 680$$

4. STATEMENT-1 : If $\log(x + z) + \log(x - 2y + z) = 2\log(x - z)$ then x, y, z are in H.P.

STATEMENT-2 : If p, q, r in AP and $\frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz}$, then x, y, z are in A.P.

STATEMENT-3 : If $\frac{a+b}{1-ab}, b, \frac{b+c}{1-bc}$ are in A.P. then $a, \frac{1}{b}, c$ are in H.P.

(1) T T F

(2) T F F

(3) T F T

(4) T T T

Sol. Answer (3)

Statement-1 :

$$\log(x+z) + \log(x-2y+z) = 2 \log(x-z)$$

$$\Rightarrow (x+z)(x-2y+z) = (x-z)^2$$

$$(x+z)^2 - (x-z)^2 = 2y(x+z)$$

$$4xz = 2y(x+z)$$

$$\Rightarrow y = \frac{2xz}{x+z}$$

Statement-2 :

$$\frac{a-x}{px} = \frac{a-y}{qy} = \frac{a-z}{rz} = k(ay)$$

$$\Rightarrow p = \frac{a-x}{kx}, q = \frac{a-y}{ky}, r = \frac{a-z}{kz}$$

$$\frac{a-x}{x}, \frac{a-y}{y}, \frac{a-z}{z} \text{ are in A.P.}$$

$$\Rightarrow \frac{a}{x} - 1, \frac{a}{y} - 1, \frac{a}{z} - 1 \text{ are in A.P.}$$

$$\frac{1}{x}, \frac{1}{y}, \frac{1}{z} \text{ are in A.P.}$$

$$\Rightarrow x, y, z \text{ are in H.P.}$$

Statement-3 :

$$\frac{a+b}{1-ab} + \frac{b+c}{1-bc} = 2b$$

$$\Rightarrow (a+b)(1-bc) + (b+c)(1-ab) = 2b(1-ab)(1-bc)$$

$$\Rightarrow (1+b^2)(2abc - (1a+c)) = 0$$

$$\therefore \frac{1}{b} = \frac{2ac}{a+c}, \text{ Hence } a, \frac{1}{b}, c \text{ are in H.P.}$$

5. **STATEMENT-1 :** If the arithmetic mean of $(b-c)^2$, $(c-a)^2$ and $(a-b)^2$ is same as the arithmetic mean of $(b+c-2a)^2$, $(c+a-2b)^2$ and $(a+b-2c)^2$ then a, b, c are equal.

STATEMENT-2 : If $\log x, \log y, \log z$ ($x, y, z > 1$) are in GP then $2x + \log(bx)$, $3x + \log(by)$, $4x + \log(bz)$ are in A.P.

STATEMENT-3 : If $n!$, $3 \times n!$ and $(n+1)!$ are in G.P. then $n!$, $5 \times n!$ and $(n+1)!$ are in A.P.

(1) T T F

(2) T F F

(3) T F T

(4) T T T

Sol. Answer (4)

Statement-1 :

$$\frac{(b-c)^2 + (a-b)^2 + (c-a)^2}{3} = \frac{(b+c-2a)^2 + (c+a-2b)^2 + (a+b-2c)^2}{3}$$

$$\Rightarrow (b+c-2a)^2 - (b-c)^2 + (c+a-2b)^2 - (a-b)^2 + (a+b-2c)^2 - (c-a)^2 = 0$$

$$\Rightarrow a = b = c$$

Statement-2 :

If $\log x$, $\log y$, $\log z$ are in A.P.

$$\Rightarrow \log(bx), \log(by), \log(bz) \text{ are in A.P.}$$

Statement-3 :

$n!$; $3 \times n!$ and $n + 1!$ are in G.P.

$$9(n!)^2 = n! (n + 1)!$$

$$\Rightarrow n + 1 = 9$$

$$\Rightarrow n = 8$$

$$\therefore n! = 8!$$

$$\Rightarrow 5 \times n! = 5 \times 8!; n + 1! = 9!$$

$$9! + 8! = 5 \times 9! \text{ Hence } n!, 5 \times n! \text{ and } n + 1! \text{ are in A.P}$$

SECTION - H

Aakash Challengers Questions

1. The sum of n terms of the series $\frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4} + \dots$ is

$$(1) \frac{1}{x-1} + \frac{k}{x^k-1}, k = 2^n \quad (2) \frac{1}{x-1} - \frac{k}{x^k-1}, k = 2^n \quad (3) \frac{1}{x-1} + \frac{k}{x^{k-1}}, k = 2^n \quad (4) \frac{1}{x-1} - \frac{k}{x^{k-1}}, k = 2^n$$

Sol. Answer (2)

After adding $\frac{1}{x-1}$ and subtracting it from given series, we get

$$\begin{aligned} S &= \frac{1}{x-1} + \frac{1}{1-x} + \frac{1}{1+x} + \frac{2}{1+x^2} + \frac{4}{1+x^4} + \dots \\ &= \frac{1}{x-1} + \frac{2}{1-x^2} + \frac{2}{1+x^2} + \frac{4}{1+x^4} + \dots \\ &= \frac{1}{x-1} + \frac{4}{1-x^4} + \frac{4}{1+x^4} + \dots \\ &= \frac{1}{x-1} - \frac{K}{x^K-1} \quad K = 2^n \end{aligned}$$

2. The first term of a sequence is 1, the second is 2 and every term is the sum of the two preceding terms. The n^{th} term is.

$$(1) \frac{1}{2^{n+1}\sqrt{5}}[(1+\sqrt{5})^{n+1} + (1-\sqrt{5})^{n+1}]$$

$$(2) \frac{1}{2^{n+1}\sqrt{5}}[(1+\sqrt{5})^{n+1} - (1-\sqrt{5})^{n+1}]$$

$$(3) \frac{1}{2^{n+1}\sqrt{5}}[(1-\sqrt{5})^{n+1} + (1-\sqrt{5})^{n-1}]$$

$$(4) \frac{1}{2^{n-1}\sqrt{5}}[(1+\sqrt{5})^{n-1} + (1-\sqrt{5})^{n-1}]$$

Sol. Answer (2)

Denoting the sequence by $a_1, a_2, \dots$ we have

$$a_n - a_{n-1} - a_{n-2} = 0 \text{ and } a_1 = 1, a_2 = 2$$

$$\text{Hence, } a_n = A\alpha^n + B\beta^n$$

$$2 = A\alpha^2 + B\beta^2$$

$$1 = A\alpha + B\beta \quad \text{where } \alpha, \beta \text{ are roots of}$$

$$\text{the equation } x^2 - x - 1 = 0$$

eliminating A, B from these equations we have

$$(\alpha - \beta) a_n = (2 - \beta)\alpha^{n-1} - (2 - \alpha)\beta^{n-1}$$

$$\text{Now, } \alpha + \beta = 1, \alpha\beta = -1$$

$$\alpha - \beta = \sqrt{5}$$

$$\therefore a_n = \frac{1}{\sqrt{5}}(\alpha^{n+1} - \beta^{n+1})$$

$$= \frac{1}{2^{n+1}\sqrt{5}}\{(1+\sqrt{5})^{n+1} - (1-\sqrt{5})^{n+1}\}$$

3. Sum of series $1 + 2x + 7x^2 + 20x^3 + \dots$ up to n terms when $x = -1$ is

$$(1) \frac{1}{16}\{4n+3+(-3)^{n+1}\} \quad (2) \frac{1}{16}\{4n-3+(-3)^{n+1}\} \quad (3) \frac{1}{16}\{4n+3+3^{n+1}\} \quad (4) \frac{1}{16}\{4n+3-3^{n+1}\}$$

Sol. Answer (1)

The given series is of the form

$$u_0 + u_1n + u_2n^2 + \dots$$

and let n^{th} term satisfy the relation

$$u_n + pu_{n-1} + qu_{n-2} = 0$$

$$\text{then } 7 + 2p + q = 0 \text{ and } 20 + 7p + 2q = 0$$

$$\therefore p = -2, q = -3$$

So, we have the relation

$$u_n - 2u_{n-1} - 3u_{n-2} = 0$$

The roots of $x^2 - 2x - 3 = 0$ are 3, -1

Therefore,

$$u_n = A \cdot 3^n + B(-1)^n$$

$$A + B = 1, 3A - B = 2 \Rightarrow A = \frac{3}{4}, B = \frac{1}{4}$$

Thus the n^{th} term $= u_{n-1}x^{n-1} = \frac{1}{4}[3^n + (-1)^{n-1}]x^{n-1}$

$$\therefore \text{Sum to } n \text{ term} = \frac{3}{4} \cdot \frac{1-3^n x^n}{1-3x} + \frac{1}{4} \cdot \frac{1-(-x)^n}{1+x}$$

$$\lim_{n \rightarrow -1} \frac{1-(-n)^n}{1+n} = n$$

$\therefore$ Sum to n terms of the series $1 - 2 + 7 - 20 + \dots$ is

$$= \frac{3}{4} \cdot \frac{1-(-3)^n}{4} + \frac{n}{4} = \frac{1}{16}[4n+3+(-3)^{n+1}]$$

4. The value of $\sum_{n=1}^{\infty} \frac{(n^2+1)}{(n+2)n!}$ is

(1) $9 - e$

(2) $\frac{9}{2} - e$

(3) $\frac{9}{2} + e$

(4) $9 + e$

Sol. Answer (2)

$$\text{Let } S = \sum_{n=1}^{\infty} \frac{n^2+1}{(n+2)(n!)} = \sum_{n=1}^{\infty} \frac{(n^2+1)(n+1)}{(n+2)!}$$

$$\text{Now, } (n^2+1)(n+1)$$

$$= n^3 + n^2 + n + 1$$

$$= a + b(n+2) + c(n+2)(n+1) + d(n+2)(n+1)(n)$$

$$a = -5, b = 5, c = -2, d = 1$$

$$\text{Thus, } S = -5 \sum_{n=1}^{\infty} \frac{1}{(n+2)!} + 5 \sum_{n=1}^{\infty} \frac{1}{n+1!} - 2 \sum_{n=1}^{\infty} \frac{1}{n!} + \sum_{n=1}^{\infty} \frac{1}{(n-1)!}$$

$$\therefore S = -5 \left(e - \frac{5}{2} \right) + 5(e-2) - 2(e-1) + e$$

$$= -5e + 5e - 2e + e + \frac{25}{2} - 10 + 2$$

$$= \frac{9}{2} - e$$

5. The numbers $x_1, x_2, x_3, \dots$ form an infinite decreasing G.P. If $x = 1$, then the common ratio of the progression for which the expression $6x_5 - 16x_4 - 3x_3 + 12x_2$ is maximum, is

- (1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) $\frac{3}{4}$ (4) $\frac{1}{2}$

Sol. Answer (4)

Let r be the common ratio then $x_2 = r, x_3 = r^2, \dots$

Also as G.P. is infinite and decreasing,

$$-1 < r < 1, r > 0 \text{ so, } 0 < r < 1 \quad \dots(i)$$

$$\begin{aligned} \text{Now } 6x_5 - 16x_4 - 3x_3 + 12x_2 \\ = 6r^4 - 16r^3 - 3r^2 + 12r = f(r) \end{aligned}$$

$$\begin{aligned} f'(r) &= 24r^3 - 48r^2 - 6r + 12 \\ &= 6[4r^3 - 8r^2 - r + 2] \\ &= 6(2r - 1)(2r + 1)(r - 2) \end{aligned}$$

$$\therefore f'(r) = 0 \text{ gives } r = \frac{1}{2}, -\frac{1}{2}, 2$$

Within the domain specified by (i) $r = \frac{1}{2}$ is the only admissible value r .

$$f''(r) = 6[12r^2 - 16r - 11]$$

$$f''\left(\frac{1}{2}\right) = 6[3 - 8 - 1] < 0$$

So $f(r)$ is maximum at $r = \frac{1}{2}$

Hence required common ratio $= \frac{1}{2}$

6. If total number of runs scored in n matches is $\frac{n+1}{4}(2^{n+1} - n - 2)$ where $n \geq 1$, and the runs scored in the k^{th} match is given by $k \cdot 2^{n+1-k}$, where $1 \leq k \leq n$, n is

- (1) 8 (2) 9 (3) 7 (4) 6

Sol. Answer (3)

$$\begin{aligned} S_n &= \sum_{k=1}^n K \cdot 2^{n+1-k} = 2^{n+1} \sum K \cdot 2^{-k} \\ &= 2^{n+2} \left[1 - \frac{1}{2^n} - \frac{n}{2^{n+1}} \right] = 2^{n+2} - 4 - 2n \end{aligned}$$

$$\Rightarrow 2^{n+2} - 2n - 4 = \frac{8}{4}(2^{n+1} - n - 2) \quad \dots(i)$$

$$\text{But } S_n = \frac{n+1}{4}(2^{n+1} - n - 2) \quad \dots(ii)$$

$$\text{From (i) and (ii) } n + 1 = 8 \Rightarrow n = 7$$

7. The sum S_n where $T_n = (-1)^n \frac{n^2 + n + 1}{n!}$ is

(1) $\frac{(-1)^n}{n-1!} - \frac{(-1)^n}{n+1!}$ (2) $\left[\frac{(-1)^n}{n!} - \frac{(-1)^n}{n-1!} \right] - 1$ (3) $\left[\frac{(-1)^{n+1}}{n+1!} - \frac{(-1)^{n-1}}{(n-1)!} \right]$ (4) $\left[\frac{(-1)^n}{n!} - \frac{(-1)^{n-1}}{(n-1)!} \right] - 1$

Sol. Answer (4)

$$T_r = (-1)^r \frac{r^2 + r + 1}{r!} \Rightarrow (-1)^r \left[\frac{r}{r-1!} + \frac{1}{(r-1)!} + \frac{1}{r!} \right]$$

$$= \left[\frac{(-1)^r}{r!} + \frac{(-1)^r}{r-1!} \right] + \left[\frac{(-1)^r}{r-1!} - \frac{(-1)^r}{(r-2)!} \right]$$

$$S_n = \sum_{r=1}^n \left[\frac{(-1)^r}{r!} - \frac{(-1)^{r-1}}{(r-1)!} \right] - \left[\frac{(-1)^{r-1}}{(r-1)!} - \frac{(-1)^{r-2}}{r-2!} \right]$$

$$= \left[\frac{(-1)^n}{n!} - \frac{(-1)^{n-1}}{(n-1)!} \right] - 1$$

8. Find a three digit number if its digits form a geometric progression and the digits of the number which is smaller by 400 form an A.P. is

(1) 139 (2) 248 (3) 842 (4) 931

Sol. Answer (4)

Let the digits be a, ar, ar^2 , then the number is $100a + 10ar + ar^2$

It is given $100a + 10ar + ar^2 - 400$

$100(a-4), ar, ar^2$ are in A.P.

$$2ar = ar^2 + (a-4)$$

$$\therefore a(1-r)^2 = 4 \Rightarrow (1-r)^2 = \frac{4}{a}$$

$1-r$ is a rational number so $\sqrt{\frac{4}{a}} = \frac{2}{\sqrt{a}}$ is rational

Since $a > 4$, $a = 9$

$$\text{Hence } 1-r = \frac{2}{3} \Rightarrow r = \frac{1}{3}$$

So digits 9, 3, 1. Hence the number 931

9. Let $S_n, n = 1, 2, 3, \dots$ be the sum of infinite geometric series whose first term is n and the common ratio is

$$\frac{1}{n+1}. \text{ Then } \lim_{n \rightarrow \infty} \frac{S_1 S_n + S_2 S_{n-1} + S_3 S_{n-2} + \dots + S_n S_1}{S_1^2 + S_2^2 + \dots + S_n^2} \text{ is}$$

(1) $\frac{1}{2}$ (2) $\frac{3}{4}$ (3) 1 (4) $-\frac{1}{2}$

Sol. Answer (1)

$$S_n = \frac{n}{1 - \frac{1}{n+1}} \Rightarrow S_n = n+1$$

$$S_1 S_n + S_2 S_{n-1} + S_3 S_{n-2} + \dots + S_n S_1 \\ = 2.(n+1) + 3.n + 4.(n-1) + \dots + (n+1)2$$

$$= \sum_{r=1}^n (r+1)(n-r+2) = \frac{n}{6}(n^2 + 9n + 14)$$

$$S_1^2 + S_2^2 + S_3^2 + \dots + S_n^2 = 2^2 + 3^2 + \dots + (n+1)^2$$

$$= \frac{(n+1)(n+2)(n+3)}{6} - 1$$

$$= \frac{n(2n^2 + 9n + 13)}{6}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{S_1 S_n + S_2 S_{n-1} + \dots + S_n S_1}{S_1^2 + S_2^2 + \dots + S_n^2} = \lim_{n \rightarrow \infty} \frac{n^2 + 9n + 14}{2n^2 + 9n + 13} = \frac{1}{2}$$

10. Let $x, y \in (0, 1)$ such that there exists a positive number $a (\neq 1)$ satisfying $\log_x a + \log_y a = 4 \log_{xy} a$.

Then the value of $\left(\frac{x}{y}\right)^{\left(\frac{y}{x}\right)}$ is....

Sol. $\log_x a + \log_y a = 4 \log_{xy} a$

$$\Rightarrow \frac{\log a}{\log x} + \frac{\log a}{\log y} = \frac{4 \log a}{\log xy}$$

$$\Rightarrow \frac{1}{\log x} + \frac{1}{\log y} = \frac{4}{\log x + \log y}$$

$$\Rightarrow \frac{\log x + \log y}{(\log x) \cdot (\log y)} = \frac{4}{(\log x + \log y)}$$

$$\Rightarrow (\log x + \log y)^2 = 4 \log x \cdot \log y$$

$$\Rightarrow (\log x - \log y)^2 = 0$$

$$\Rightarrow \log x = \log y \Rightarrow x = y \Rightarrow \frac{x}{y} = 1$$

$$\Rightarrow \left(\frac{x}{y}\right)^{\frac{y}{x}} = (1)^1 = 1$$

11. Let $a = \frac{x}{1+x^2} + \frac{1}{3}\left(\frac{x}{1+x^2}\right)^3 + \frac{1}{5}\left(\frac{x}{1+x^2}\right)^5 + \dots$ and $b = x - \frac{2}{3}x^3 + \frac{1}{5}x^5 + \frac{1}{7}x^7 - \frac{2}{9}x^9 \dots$ then

(1) $a = 2b$

(2) $a = b$

(3) $2a = b$

(4) $2a = 3b$

Sol. Answer (2)

$$a = \frac{1}{2} \log \left[\frac{1 + \frac{x}{1+x^2}}{1 - \frac{x}{1+x^2}} \right] = \frac{1}{2} \log \left(\frac{1+x+x^2}{1-x+x^2} \right)$$

$$\begin{aligned} b &= \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \right) - \left(x^3 + \frac{x^9}{3} + \frac{x^{15}}{5} + \dots \right) \\ &= \frac{1}{2} \log \left(\frac{1+x}{1-x} \right) - \frac{1}{2} \log \left(\frac{1+x^3}{1-x^3} \right) \\ &= \frac{1}{2} \log \left(\frac{1+x+x^2}{1-x+x^2} \right) \end{aligned}$$

Hence $a = b$

12. $\frac{1}{1!} + \frac{1+3}{2!}x + \frac{1+3+5}{3!}x^2 \dots$

(1) $e^x(1+x)$

(2) $e^x(1-x)$

(3) xe^x

(4) $e^x(x+2)$

Sol. Answer (1)

$$\begin{aligned} T_n &= \frac{n^2}{n!} x^{n-1} = \frac{n^2}{n(n-1)!} x^{n-1} = \frac{(n-1+1)}{n-1!} x^{n-1} \\ &= \left[\frac{1}{(n-2)!} + \frac{1}{(n-1)!} \right] x^{n-1} \Rightarrow T_n = \frac{x^{n-1}}{n-2!} + \frac{x^{n-1}}{n-1!} \end{aligned}$$

$$S = T_1 + T_2 + T_3 + \dots$$

$$\begin{aligned} S &= x \left[1 + \frac{x}{1!} + \frac{x^2}{2!} \dots \right] + \left[1 + \frac{x}{1!} + \frac{x^2}{2!} \dots \right] \\ &= xe^x + e^x = e^x(x+1) \end{aligned}$$

13. Let 2048 arithmetic means be inserted between $2^{22} - 1$ and $2^{22} + 1$. Suppose the sum of these arithmetic means is S . Then $\frac{S}{2^{23}}$ is ...

Sol. Sum of n , A.M.'s is n times the single A.M.

$$S = \frac{2048[2^{22} - 1 + 2^{22} + 1]}{2}$$

$$S = 1024 [2 \cdot 2^{22}] = 1024 \cdot 2^{23}$$

$$\frac{S}{2^{23}} = 1024$$

14. Let $S = \frac{2+6}{4^{100}} + \frac{2+2 \times 6}{4^{99}} + \frac{2+3 \times 6}{4^{98}} + \dots + \frac{2+99 \times 6}{4^2} + \frac{2+100 \times 6}{4}$. Then $3S$ equals.

Sol. $S = 2 \left[\frac{1}{4^{100}} + \frac{1}{4^{99}} + \dots + \frac{1}{4} \right] + 6 \left[\frac{1}{4^{100}} + \frac{2}{4^{99}} + \frac{3}{4^{98}} + \dots + \frac{100}{4} \right]$

$$\Rightarrow S = 2S_1 + 6S_2$$

Where $S_1 = \frac{1}{4^{100}} + \frac{1}{4^{99}} + \dots + \frac{1}{4}$ and $S_2 = \frac{1}{4^{100}} + \frac{2}{4^{99}} + \frac{3}{4^{98}} + \dots + \frac{100}{4}$

$$\begin{aligned} S_1 &= \frac{1}{4^{100}} + \frac{1}{4^{99}} + \dots + \frac{1}{4} \\ &= \frac{1}{4} + \frac{1}{4^2} + \dots + \frac{1}{4^{100}} \\ &= \frac{\frac{1}{4} \left(1 - \left(\frac{1}{4} \right)^{100} \right)}{\left(1 - \frac{1}{4} \right)} = \frac{1}{3} \left(1 - \left(\frac{1}{4} \right)^{100} \right) \end{aligned}$$

$$S_1 = \frac{1}{4^{100}} + \frac{2}{4^{99}} + \frac{3}{4^{98}} + \dots + \frac{100}{4} \quad \dots (i)$$

$$S_2 \times 4 = \frac{1}{4^{99}} + \frac{2}{4^{98}} + \dots + \frac{99}{4} + 100 \quad \dots (ii)$$

Subtracting

$$S_2 - 4S_2 = \frac{1}{4^{100}} + \frac{1}{4^{99}} + \dots + \frac{1}{4} - 100$$

$$-3S_2 = \left(\frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \dots + \frac{1}{4^{100}} \right) - 100$$

$$-3S_2 = \frac{\frac{1}{4} \left(1 - \left(\frac{1}{4} \right)^{100} \right)}{\left(1 - \frac{1}{4} \right)} - 100$$

$$-3S_2 = \frac{1}{3} \left(1 - \left(\frac{1}{4} \right)^{100} \right) - 100$$

$$-3S_2 = -\frac{1}{3} \left(1 - \left(\frac{1}{4} \right)^{100} \right) + 100$$

Now,

$$S = 2S_1 + 6S_2$$

$$= 2 \left(\frac{1}{3} \left(1 - \left(\frac{1}{4} \right)^{100} \right) \right) + 2 \left(-\frac{1}{3} \left(1 - \left(\frac{1}{4} \right)^{100} \right) \right) + 200$$

$$S = 200$$

$$\Rightarrow 3S = 600$$

15. The sum of $\sum_{n=1}^{16} \frac{n^4}{4n^2-1}$ is

$$\begin{aligned} \text{Sol. } t_n &= \frac{n^4}{4n^2-1} = \frac{16n^4-1+1}{16(4n^2-1)} = \frac{1}{16}(4n^2+1) + \frac{1}{16(4n^2-1)} \\ &= \frac{1}{16}(4n^2+1) + \frac{1}{32} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right) \\ \sum_{n=1}^{16} t_n &= \frac{1}{16} \left[\frac{4 \cdot 16 \cdot 17 \cdot 33}{6} + 16 \right] + \frac{1}{32} \left(1 - \frac{1}{33} \right) = 375 + \frac{1}{33} = \frac{12376}{33} \end{aligned}$$

16. Four different integers form an increasing AP. One of these numbers equal to the sum of squares of the other three numbers. Find the numbers.

Sol. Let the numbers be $a-d, a, a+d, a+2d$

Since this is an increasing A.P. of integers, $a, b \in \mathbb{Z}$ and $d > 0$

Hence $(a-d)^2 + a^2 + (a+d)^2 = a+2d$

i.e., $2d^2 - 2d + 3a^2 - a = 0$

$$\begin{aligned} \therefore d &= \frac{1}{4} \left[2 \pm \sqrt{4 - 8(3a^2 - a)} \right] \\ &= \frac{1}{2} \left[1 \pm \sqrt{1 + 2a - 6a^2} \right] \end{aligned}$$

Since d is positive integer

$1 + 2a - 6a^2 > 0$ and perfect square

Now $6a^2 - 2a - 1 < 0$

$$\left(a - \frac{1}{6} \right)^2 - \frac{1}{36} - \frac{1}{6} < 0$$

$$\text{Or, } \left(a - \frac{1}{6} \right)^2 < \left(\frac{\sqrt{7}}{6} \right)^2$$

$$-\frac{\sqrt{7}}{6} < a - \frac{1}{6} < \frac{\sqrt{7}}{6} \text{ since } a \text{ is integer,}$$

$a = 0$, then $1 + 2a - 6a^2 = 1$ which is a perfect square

then $d = \frac{1}{2}(1 \pm 1) = 1$ or 0 since $d > 0$

$$\therefore d = 1$$

Hence the number $-1, 0, 1, 2$

17. $\frac{\sum_{r=1}^n r^4}{\sum_{r=1}^n r^2}$ is equal to

Sol. $(x+1)^5 - x^5 = 5x^4 + 10x^3 + 10x^2 + 5x + 1$

$$\Sigma[(x+1)^5 - x^5] = 5\Sigma x^4 + 10\Sigma x^3 + 10\Sigma x^2 + 5\Sigma x + \Sigma 1$$

Where Σ stands for $\sum_{x=1}^n$

$$\therefore (x-1)^{-1} = 5\Sigma x^4 + \frac{10n^2(n+1)^2}{4} + \frac{10n(n+1)(2n+1)}{6} + \frac{5n(n+1)}{2} + n$$

$$5\Sigma x^4 = n^5 + 5n^4 + 10n^3 + 10n^2 + 5n - \frac{5}{2}n^2(n+1)^2 - \frac{5}{3}n(n+1)(2n+1)$$

$$- \frac{5n(n+1)}{2} - n$$

$$= \frac{n(n+1)}{6}(6n^3 + 9n^2 + n - 1)$$

$$= \frac{n(n+1)(2n+1)(3n^2 + 3n - 1)}{6}$$

$$\therefore \frac{\Sigma x^4}{\Sigma x^2} = \frac{3n(n+1) - 1}{5}$$

18. $a_n = \frac{3}{4} - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots + (-1)^n \left(\frac{3}{4}\right)^n$ and $b_n = 1 - a_n$, then find the last natural number n_0 such that $b_n > a_n \forall n \geq n_0$.

Sol. $a_n = \frac{3}{4} - \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots + (-1)^n \left(\frac{3}{4}\right)^n = \frac{\frac{3}{4} \left(1 - \left(-\frac{3}{4}\right)^n\right)}{1 + \frac{3}{4}}$

$$= \frac{3}{7} \left[1 - \left(-\frac{3}{4}\right)^n\right] \text{ Since } b_n = 1 - a_n \text{ and } b_n > a_n \forall n \geq n_0$$

$$\Rightarrow \frac{6}{7} \left(1 - \left(-\frac{3}{4}\right)^n\right) < 1$$

$$\Rightarrow (-3)^{n+1} < 2^{2n-1}$$

When n is even, inequality is held $\forall n \in \mathbb{N}$ and, for n is odd, inequality is held, when $n \geq 7$

19. If a, b, c are in AP and b, c, a are in GP, then show that c, a, b are in H.P. Find $a : b : c$

Sol. $a + c = 2b$... (i)

$$ab = c^2 \quad \dots (ii)$$

$$\frac{2bc}{b+c} = \frac{c(a+c)}{b+c} \text{ by the (i)}$$

$$= \frac{ca + ab}{b+c} = a \text{ by (ii)}$$

$\therefore c, a, b$ are in H.P.

Eliminating of a from (i) and (ii) gives

$$b(2b - c) = c^2$$

$$\Rightarrow c^2 + bc - 2b^2 = 0$$

$$(c - b)(c + 2b) = 0$$

$$\Rightarrow \frac{c}{b} = -2 \because (b \neq c) \quad \dots (iii)$$

$$a = 2b - c \text{ by (i)}$$

$$\frac{a}{b} = 2 - \frac{c}{b} = 4$$

$$\therefore a : b : c = 4 : 1 : -2$$



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